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Device Characterization

Device Characterization

Device characterization involves investigating the behavior of a DUT under various input conditions. The following topics are available:

- [Device Characterization Overview](#)
- [Characterization Modes](#)
- [Performing a Shmoo](#)
- [Performing Adjust and Measure](#)

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Device Characterization : Device Characterization Overview

Device Characterization Overview

Device characterization involves investigating the behavior of a DUT under various input conditions. This activity can be useful both during test program debug and during production testing. Certain combinations of input parameters cause a DUT to fail a test, while other combinations can cause it to pass. Exploring these pass/fail regions can provide information about the manufacturing process and the appropriateness of the chosen test limits.

Characterization Features

IG-XL provides flexibility in configuring and executing a characterization operation. You can:

- Perform characterization from a Flow Table as part of the regular testing process or interactively while stopped at a breakpoint in the IG-XL debugger.
- Adjust hardware parameters such as Vih, Vps, drive Data time, window Open time, and other parameters on the Pin Levels, Edge Sets, and Time Sets sheets.
- Adjust spec values. IG-XL determines the full effect of changing a spec value, modifying any dependent hardware values as needed.
- View the results of a characterization or send them to a file or to a DataTool worksheet.
- Use characterization setups to define what parameters vary, by what amounts, according to what algorithm, and so on.

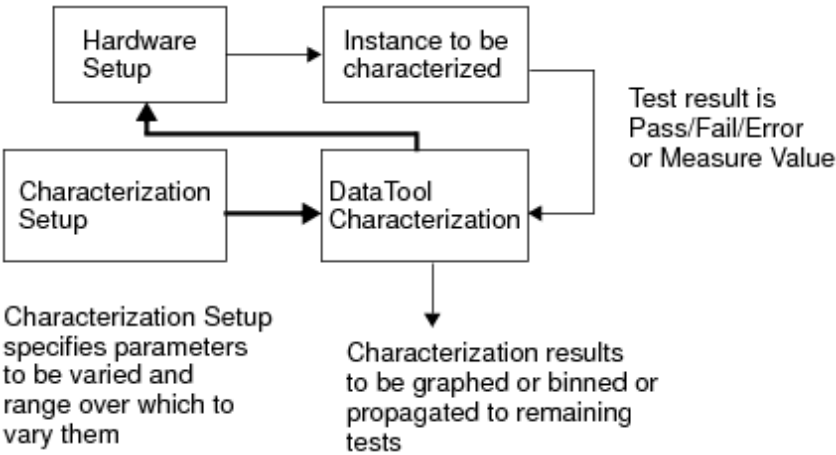
Setups are stored with the test program on the Characterization sheet. You can call them from a Flow Table as part of a test program or from a Characterization Editor for interactive debugging. You

can also use the Characterization Editor to create or modify setups on the fly. There is also a VBT interface for dynamically creating, modifying, storing, and executing setups.

Note that characterization from a Flow Table while stopped at a breakpoint has some limitations when performing concurrent testing. See Use of Characterization Studio with Concurrent Test.

Characterization and Test Instances

The following figure provides an overview of characterizing a test instance.



Characterizing a Test Instance

Characterization is always performed on a user-selected test instance. The instance can be any kind of test. Characterization does not interfere with or require any particular setup of the tester, and it is generally subject to the same IG-XL control (such as selected and active sites, datalogging, and pass/fail binning) as other test activities.

In other words, think of characterization as being like any other kind of test. Like the Test opcode, it is site-aware, so that each site is measured and compensated individually.

Characterization Setups and Steps

Each characterization is defined in a named setup, which contains the parameters for the characterization and controls the execution of the characterization operations. A setup can contain multiple steps, where each step has its own characterization mode and corresponding setup parameters.

Setups are stored on the Characterization sheet in the program workbook. Characterization Studio is an interactive tool for creating, editing, and running characterization setups and displaying the results. There is also a VBT interface for performing these operations.

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Device Characterization : Characterization Modes

Characterization Modes

IG-XL supports the following modes or types of characterization:

Shmoo (1-D, 2-D, or 3-D)

A Shmoo varies a hardware parameter or spec over a range and at each point it either determines a pass/fail result (functional characterization) or measures a parametric value (parametric characterization). A Shmoo can support up to three axes (X, Y, and Z). Each axis supports one primary parameter and multiple tracking parameters. A tracking parameter is varied in unison with the primary parameter.

Measure

A Measure varies a hardware parameter or spec over a range until it locates a user-specified transition point between pass and fail. It then usually executes a test that compares the measured value to a user-specified set of pass/fail limits. The parameters and the tester state are always returned to their pre-Measure state at the end of a Measure characterization.

Adjust

An Adjust is a Measure with an additional operation. Like a Measure, it varies a hardware parameter or spec over a range until it locates a user-specified transition point between pass and fail. In addition, the Adjust can set an output spec to the final result. The output spec can propagate the adjusted spec value for use in subsequent test instances: in this way, a spec value can be adjusted during program execution. The method of propagating the adjusts value is called an overlay. You can also apply a backoff value to the found transition point so the result is not located at an unstable transition point.

If no output spec is specified, the Adjust leaves the parameter and tester state at the transition point, including any user-specified backoff. If an output spec is specified, the result is applied to the output spec and the parameter and the tester state is restored to the pre-Adjust state.

Margin

A Margin is different from the other characterization modes. It is used to find the range over which a passing test continues to pass. A Margin varies a hardware parameter or spec over a range. For each point defined, it either rebursts the pattern or reruns the test, and records a pass/fail/error for the point. The parameters and the tester state are always returned to their pre-Margin state at the end of a Margin.

A Margin is like a 1-D Shmoo, but does not permit any tracking parameters.

	
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Device Characterization : Performing a Shmoo

Performing a Shmoo

This section provides information on running a shmoo, including:

- | | |
|---|----------------------------------|
| • | Primary and Tracking Parameters |
| • | Optimizing 2-D and 3-D Shmoos |
| • | Shmoos with Measures and Adjusts |
| • | Shmooing a Parametric Test |

No Run-Time Checking

Note that no error checking occurs when running a Shmoo. Even if an invalid condition occurs when a value is being varied (for example, if V_{ih} becomes smaller than V_{il}), a Shmoo may be created and displayed without indicating that an error has occurred.

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Device Characterization : Performing a Shmoo : Primary and Tracking Parameters

Primary and Tracking Parameters

A Shmoo varies one or more parameters over a range while recording Pass/Fail/Error information. A Shmoo can vary up to three primary parameters, which are displayed on the X, Y, and Z axes. In addition, any primary parameter can have any number of other secondary parameters that track the variation on the primary parameter.

A Shmoo setup can define primary parameters that correspond to X, Y, and Z axis display values. The X axis is the fastest axis, changing on every point of the Shmoo. The Y axis changes more slowly than the X axis, and the Z axis changes more slowly than Y. Parameters attached to the Y and Z axis should be those with the longest settling time.

Each primary parameter can have one or more tracking parameters. Tracking parameters are varied in lock step with the primary parameter. A tracking parameter uses the same number of steps as the primary parameter. All parameters for an axis are updated together before a test instance is re-executed. For example, if the X axis has three parameters, all three parameters are updated before executing the test instance.

Tracking parameters are useful when one parameter must move with another. For example, the output voltage comparator may need to track the power supply voltage. If the power supply voltage is varied over several steps for a Shmoo, the output parameter must vary by the same steps. If the comparator voltage level has a spec-based definition using the power supply, then it will vary properly for the Shmoo. But if the two definitions are independent, the Shmoo setup must explicitly define a tracking parameter. In this case, the power supply voltage is the primary parameter for that axis, and the output comparator voltage is defined as a tracking parameter on the same axis.



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Device Characterization : Performing a Shmoo : Optimizing 2-D and 3-D Shmoos

Optimizing 2-D and 3-D Shmoos

An enumerated value has been added to the set of characterization features that enables optimization of 2-D and 3-D shmoo execution. By default, the optimization is not enabled. To enable it, use this VBT statement:

```
TheExec.DevChar.Configuration.Features _  
    (tlDevCharFeature_OptimizeSpecLoadForShmoo).Enabled = True
```

The more heavily used spec should be assigned to the Y or Z axis, which loads the spec less frequently than the X axis. For example, assume a Shmoo setup contains two specs: acSpec1 in X Axis and acSpec2 in Y axis. acSpec1 is used in 100 places (it is used in a formula) in a Time Sets sheet, and acSpec2 is used in 500 places in a Time Sets sheet. In this case, it is advisable to use acSpec2 in the Y axis. This provides additional performance improvement only if the feature above is enabled.

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Device Characterization : Performing a Shmoo : Shmoos with Measures and Adjusts

Shmoos with Measures and Adjusts

A Shmoo setup can contain steps that perform a Measure or Adjust. This is similar to a regular Shmoo, except that, instead of rebursting or retesting, it performs one or more Measures at each point. In addition, you can perform an Adjust before the Measures, to leave the tester in an adjusted state. A typical application is shmooing Vdd, adjusting certain other parameters, and then measuring maximum operating frequency.

Note that, when a setup contains a Shmoo plus a Measure or Adjust, the Measure or Adjust is executed at each point of the Shmoo. That is, they are essentially executed simultaneously, not sequential: regardless of its position in the setup, the Measure or Shmoo is not executed before or after the Shmoo, but with the Shmoo. Because of this, an Adjust From cannot reference an Adjust that is part of a Shmoo setup: because the Adjust is executed at each point of the Shmoo, there is no single Adjust that the Adjust From can reference.

A Shmoo can perform one or more Measures. At each point determined by the Shmoo’s axes, the Measures are performed under the current set of conditions. Each Measure finds the pass/fail transition point and returns the measured value. In addition, if no transition point is found, the Measure returns a Fail. A Pass means that a transition point was found. If a setup contains multiple Measures, each Measure is executed independently, and the result of each is recorded.

Another possibility for a Shmoo is to have an Adjust followed by a Measure. At each point, the Adjust can find a pass/fail transition. You can then apply a “backoff” value, leaving the tester in a “just passing” state. (An Adjust restores tester state only if it applies its value to a spec.) Then the Measure is performed, returning a measured value under the current Shmoo conditions and the “just passing” state.

If a Shmoo setup also contains Adjusts or Measures, the order of operations is:

1. Apply the X, Y, and Z axis parameter values to set up the current conditions.

2. Perform any Adjusts in the order they are defined in the setup. Propagate the results to specs, if desired.
3. Perform any Measures in the order they are defined in the setup.
4. If Measures were performed, move to the next X, Y, and Z point. If no Measures were performed, either reburst the pattern or rerun the test, as specified; then move to the next X, Y, and Z point.

Note the implication of the final step. At each point, there will be either a retest/reburst (normal Shmoo), or one or more Measure operations, but not both.

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Device Characterization : Performing a Shmoo : Shmooing a Parametric Test

Shmooing a Parametric Test

Typically, you use a Shmoo to characterize a functional test, by rebursting a pattern under varying conditions. You can also characterize a parametric test. The procedure is similar to a functional test, with the differences noted below.

A parametric test generates a measured value for each pin. The measured values are compared to the test limits to determine the test’s Pass/Fail result.

The test datalogs the Pass/Fail result. IG-XL provides a way to access and manipulate the measured values programmatically. See `TheExec.Results` in the VBT Syntax Reference. If a Shmoo is run interactively, the Characterization Studio window can display the results.

Characterizing Idd versus Vdd

You can use the DCVI or DCVS PowerSupply VBT library test to create a test instance that can then be characterized. The PowerSupply tests are particularly suited to a Shmoo, because they provide a way to vary the force condition. Note these points:

- Create a test instance that uses a PowerSupply library test. Set the VoltageOutput argument to psSourceAmbient. This value causes the test to use whatever Vps value is specified by the Pin Levels sheet that is used by that instance.
- Create a characterization setup that contains a single X Shmoo step. Set Parameter Type to Level, and Parameter Name to Vps. Specify the range over which you want Vps to be varied, the number of steps, and so on.

Set the Test Method to Retest, not Reburst. This is a requirement for any setup that is to be applied to a parametric test.

- In addition, to manipulate the parametric values datalogged by the test, you can create an End interpose function.

Characterizing PPMU and Other Parametric Tests

Characterize PPMU and other parametric tests in much the same way as the PowerSupply test. However, the kinds of characterization you can perform are limited. Because these VBT library tests do not have a VoltageOutput setting, you cannot change the force condition being used by the test. For some Shmoos, this is not a serious limitation. For example, to shmoo input leakage current as a function of supply voltage, use the procedure outlined above. The characterization would vary the Vps value while calling the PinPmu test to measure and test the leakage currents of the DUT pins. Note, however, that this test does not vary the PPMU voltage being applied to the DUT pins. Instead, it varies the supply voltage to the DUT.

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Device Characterization : Performing Adjust and Measure

Performing Adjust and Measure

This topic includes the following items:

- Running an Adjust or Measure
- Troubleshooting an Adjust or Measure
- Algorithms for Adjusts and Measures
- Examples of Adjust/Measure Algorithms

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Device Characterization : Performing Adjust and Measure : Running an Adjust or Measure

Running an Adjust or Measure

This topic includes the following items:

- [Adjust Overview](#)
- [Parts of an Adjust](#)
- [Contexts of the Setup and the Overlaid Test](#)
- [Adjust and Measure Failures](#)
- [Adjust From Mode, A Collection of Setups](#)
- [Optimizing Timing Adjusts](#)

Additional information is available for these topics:

- [Troubleshooting an Adjust or Measure](#)
- [Algorithms for Adjusts and Measures](#)
- [Examples of Adjust/Measure Algorithms](#)

Note that you can run a Measure from SpecTool.

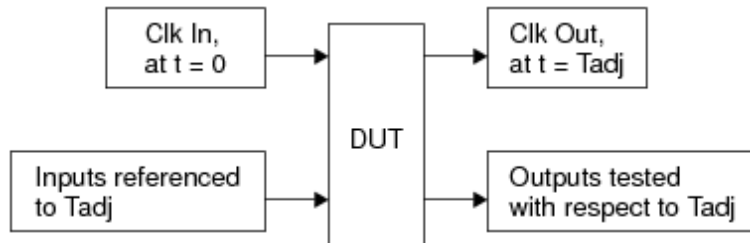
Adjust Overview

An Adjust operation varies a hardware parameter or spec over a range until the transition point between Pass and Fail is located. An Adjust is sometimes known as an “edge find.”

Adjust is closely related to Measure. The difference is that an Adjust determines a value and then propagates it to other sheets by means of a spec. In contrast, Measure determines the value without propagating it.

The two primary uses of performing an Adjust in a test program are:

- To test a propagation delay or input voltage threshold.
- To locate an output edge of the DUT and then adjust the tester setup based on the location of that edge. This is done in cases where the DUT timing is referenced to an output event that cannot be determined before run time. The following illustration shows an example of this use.



Locating a DUT Output Edge

Once T_{adj} is measured, an AC spec can be used to propagate its value to timing sheets, causing the recalculation of edge and window placements for the device.

Parts of an Adjust

The Adjust setup specifies the parameter to be varied and the range over which it should be varied, plus the algorithm to be used to find the Pass/Fail transition point: either a variation on binary search, or a linear step method. See Algorithms for Adjusts and Measures and Examples of Adjust/Measure Algorithms.

The setup also specifies a levels or timing spec that is to receive the parameter value found by the Adjust. It is assumed that this spec will be used in the definition of other levels or timing values. In this way, the value found by the Adjust can be propagated to the rest of the test program.

The Adjust operation causes the test to be repeatedly performed while it varies the setup conditions. The process is complete when the test just barely passes.

If the adjusted value is not applied to a spec, the Adjust leaves the tester in the “just passing” state.

If the adjusted value is applied to a spec, the tester hardware state is restored.

The parameter value in force at the end of the Adjust can then be applied to subsequent test instances in the flow. If the setup has specified a “backoff” value, this value is applied to the result to leave the test passing with a certain safety margin.

If a subsequent test instance has an overlay that specifies the Adjust setup, the Adjusted value is overlaid on top of that base timing or levels definition.

The adjusted values generated by a setup remain available from one device to the next, as long as the program is running, unless the values are overwritten by a subsequent execution of the same setup. You can use this feature to improve test times. See [Optimizing Timing Adjusts](#).

Contexts of the Setup and the Overlaid Test

Each test instance is executed within the context that the instance has specified:

- | |
|---|
| <ul style="list-style-type: none">• An AC context, consisting of a Time Sets sheet and AC category and selector |
| <ul style="list-style-type: none">• A DC context, consisting of a Pin Levels sheet and DC category and selector |

When you run an Adjust, the test that is characterized using a setup has a context, and the test on which the adjusted value is overlaid also has a context, perhaps a different one.

IG-XL allows you to run a setup in one context and to apply the adjusted value to a test with a different context. When a setup is run, IG-XL finds all the tests on which the adjusted value will be overlaid, and it creates an overlay specifically for the context of each of the tests.

Use this feature to run a slow speed test to gather some delay or setup/hold values that are independent of clock rate. These values can then be overlaid on tests that are run at a variety of speeds.

Adjust and Measure Failures

If an Adjust or a Measure fails, the following error messages can provide additional information:

"Stuck at Pass"/"Stuck at Fail"	An Adjust or Measure locates the point of pass/fail transition. If the test pass or fails at all points between From and To, there is no transition, and the operation fails. "Stuck at Pass" means that the test passed at all points. "Stuck at Fail" means it failed at all points. In addition, the Pos Linear method is stuck at pass if the first point in the search range is a Pass, and Neg Linear is stuck at fail if the first point is a Fail. See Algorithms for Adjusts and Measures. See also Examples of Adjust/Measure Algorithms.
"Busy Adjusting"	The Adjust or Measure has been stopped, but the adjust process is incomplete. This can be caused by either of the following: • A linear search has been specified and the maximum iteration count of 100 tests has been reached.

- An Adjust/Measure was started but all sites are inactive, perhaps because failed tests caused parts to be binned out. This means the Adjust/Measure has no device on which to perform its operation.

On failures, see [Troubleshooting an Adjust or Measure](#).

[Adjust From Mode, A Collection of Setups](#)

A standard Adjust consists of a single setup, containing one or more Adjust operations, that characterizes a single test instance. The result of each Adjust is written to a spec, whose value can then be overlaid on a later test instance.

You may want to have specs whose adjusted values come from different setups that have been applied to different test instances. For example, SetupA used with TestB may adjust SpecC, and Setup1 used with Test2 may adjust Spec3. You now want to overlay both the adjusted specs SpecC and Spec3 on TestX, but only one setup can be overlaid on a test instance.

The solution is to use a setup with steps that specify the mode Adjust From. This mode does not perform an Adjust. Instead, it takes the value of the spec named as the Adjust Spec Name, as it has been adjusted by the previous execution of the setup named as the Adjust From Setup. In this example, a SetupZ can be created, with two steps: one using SpecC as adjusted by SetupA, and one using Spec3 as adjusted by Setup 1. SetupZ can then be used as the argument to a characterize opcode on the Flow Table. Then SetupZ can be overlaid on a subsequent test instance, with the result that the test instance uses the adjusted values of SpecC and Spec3.

A setup that contains Adjust From steps can also contain Adjust or other operations.

Note that an Adjust From cannot refer to an Adjust that is in the same setup as a Shmoo. If a setup has a Shmoo and an Adjust, the Adjust is executed at each point of the Shmoo. This means there is no single Adjust result that the Adjust From can reference. See [Shmoos with Measures and Adjusts](#).

[Optimizing Timing Adjusts](#)

You can perform a timing Adjust on every device tested, but it is faster to perform the Adjust only if the device is operating at a point that is sufficiently different to warrant it. Because the overlay values generated by an Adjust persist for as long as the program is executing, the adjusted values generated by characterizing one device can be used for subsequent devices. This permits an optimizing approach.

The approach is to insert a new verification test before the test that performs the Adjust. The resulting sequence is as follows:

- The verification test, whose definition on a Test Instances sheet has an Overlay column that references the setup for the Adjust test.

- The Adjust test, whose entry in a Flow Table has a group or device qualifier that executes this test only if the verification test fails.

On the first execution of the test program, there is no overlay value, and the verification test uses the timing values loaded into hardware. On subsequent executions, the Adjust has been performed, and the verification test uses the overlay value. The test can verify that the device timing is good enough to use in the rest of the test program:

- If the verification test passes, the Adjust test is not executed, and the existing overlay value is used.
- If the verification test fails, the Adjust test is executed, and a new overlay value is generated.

In this way, the time-consuming timing Adjust is executed only when necessary.

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Device Characterization : Performing Adjust and Measure : Troubleshooting an Adjust or Measure

Troubleshooting an Adjust or Measure

Lists in this topic provide information for debugging or troubleshooting an Adjust or a Measure.

Debugging an Adjust or Measure

Use the following techniques to debug Adjusts and Measures in IG-XL:

- A test instance to be characterized must be stable. It must never fail because of marginal timing or intermittent sync-up. To verify that you have a stable test that can be used as the basis of an Adjust or Measure, do the following:

a. Create a stand-alone test of the instance.

b. Use the debugger to break at the test.

c. Bring up the Pattern Control debugger and loop on the test until it fails. If it fails, use standard debugging techniques to find the problem.
 - Modify the setup to call an PostStep interpose function of your own creation. Set a breakpoint in the interpose function and execute the test program in the IG-XL debugger. Once you are at the breakpoint, use the regular debug displays or use live VBT commands to interrogate the state of the hardware at the completion of the adjust/measure.
 - Similar to the preceding technique, create a PostPoint interpose function that is called by the instance being characterized. Use this to examine the state of the tester during the course of measuring the value.

- The value of a measured/adjusted result should not depend on the algorithm type used. Try duplicating the setup step and modifying it to use another algorithm. For example, you may have performed an Adjust using the Successive algorithm. Add a second Adjust to the same setup and try using the Binary algorithm. Add a third and try the Linear algorithm. All algorithms should return the same answer. If the results differ, you have a drifting or unstable device behavior or a nonrepeatable test instance; or you have multiple events occurring in the search window.

Reasons an Adjust or Measure Can Fail or Give an Incorrect Answer

An Adjust or Measure can fail or give incorrect answers for the following reasons:

- The DUT requires sync-up, but the pattern does not perform sync-up, or the sync-up is unstable.

This causes unrepeatable burst behavior, confusing the adjust algorithm. Note that the adjuster usually finds an answer, but it is unstable and incorrect.

- The setup does not focus its actions on a pin list or time set list.

The setup applies values to all pins in all time sets, creating illegal conditions for the DUT.

- The search range is too broad, encompassing multiple pin events.

The adjust algorithm assumes a single transition point in the search range.

- If the setup is modifying an Edge Open or Close time, the pins being Applied To must be in Compare Edge or Window mode.

Compare mode Off causes Open/Close timing changes to have no effect.

- If doing multiple adjusts/measures in different tsets in the same pattern, be sure that the pattern passes when run stand-alone. If it does not pass it means that the adjuster cannot influence the burst's pass/fail result when it varies its parameter. The result is a "stuck at" condition.

Speeding Up Adjust or Measure

Use the following suggestions to speed up an Adjust or Measure:

- Use the Binary algorithm rather than Successive.
- Use the Linear algorithm with large step sizes.
- Narrow the search range or increase the desired resolution or both.
- Conditionally perform Adjust.

Overlays have lifetimes that extend past the testing of a single device. Add an overlaid test in your Flow Table to check whether that overlay's current values are adequate. If the test fails, reperform the Adjust to create new values for the overlay.

		
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Device Characterization : Performing Adjust and Measure : Algorithms for Adjusts and Measures

Algorithms for Adjusts and Measures

A characterization setup requires a value for the Point Generation Algorithm to specify the method to search for the pass/fail transition.

For an Adjust or Measure, the options are:

Binary	Generate values by continuously selecting the midpoint of the defined range (From and To) and updating the endpoints of the range. The algorithm continues updating the endpoints until the range is equally to or less than the step size. This algorithm produces values that are integer increments of the step size, with values rounded up if necessary.
Linear	Generate values from From to To in fixed steps calculated from Steps and Step Size. This algorithm produces values up to and including the end of the range but will not exceed the end of the range.
Successive	Generate values using a combination of binary and linear search algorithms. A binary search is used until a transition is found between two consecutive points, and then a linear search is used between these points. Use the Successive algorithm with characterization of a response that drifts with time.
Jump	Generate values using a linear search that starts with the From value and skips the number of points specified by the Arguments field.

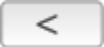
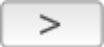
For more information, see Examples of Adjust/Measure Algorithms.

Notes on the Algorithms

Note these points on the algorithms:

- If the From value is greater than the To value, the search is a backwards search, from the higher value (From) to the lower value (To).
- For the search to succeed, the pass/fail condition must fall with the search range. Otherwise, the search fails, giving a "Stuck at Pass" or "Stuck at Fail" error.
- A Linear search will fail if the first point it finds is a Pass (for FailToPass transition) or a Fail (for FailToPass transition). The search gives a "Stuck at Pass" or "Stuck at Fail" error. To make this check before running the setup, set ResultsCheck to First, which checks whether the first point in the range satisfies the desired transition.
- The binary searches assume a single transition point; these searches stop as soon as they find a transition. If a range has more than one transition point, use a linear search to find them.
- HPM and UltraPin4000 restriction:

There is a restriction on the algorithms that can be used to characterize the drive edges of site-copied channels simultaneously, that is, in parallel. This situation can occur when a pin on the channel map is assigned site-copied channels on different sites. For this characterization, you can use only the Linear or List algorithm. Any other algorithm may fail to find the transition point.

		
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CZConcepts.2.13

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Device Characterization : Performing Adjust and Measure : Examples of Adjust/Measure Algorithms

Examples of Adjust/Measure Algorithms

The following examples show in detail exactly how the Adjust/Measure algorithms work. Each example shows, for the transition depicted, how many steps each algorithm takes to succeed in finding the transition point or to get stuck at a pass or fail.

In all cases, the Successive algorithm is the most robust and is guaranteed to succeed, if possible. The examples, however, suggest that, in certain cases, a different algorithm may be more efficient in that it takes fewer steps to find the transition point.

The following examples are from 0.00 to 1.00:

- | | |
|--|---|
| | <ul style="list-style-type: none">Fail to Pass Transition at 0.61 |
| | <ul style="list-style-type: none">Pass To Fail Transition at 0.61 |
| | <ul style="list-style-type: none">Fail To Pass Transition at 1.1 |
| | <ul style="list-style-type: none">Pass To Fail Transition at 1.1 |

The following examples are from 1.00 to 0.00:

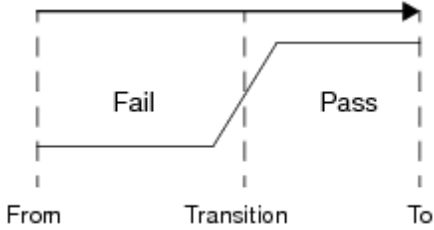
- | | |
|--|---|
| | <ul style="list-style-type: none">Fail to Pass Transition at 0.61 |
| | <ul style="list-style-type: none">Pass To Fail Transition at 0.61 |
| | <ul style="list-style-type: none">Fail To Pass Transition at 1.1 |

• Pass To Fail Transition at 1.1

For more information, see [Algorithms for Adjusts and Measures](#).

Fail to Pass Transition at 0.61

From: 0.00, To: 1.00, Step: 0.05



Algorithm = Successive

Transition = First	Transition = PassToFail	Transition = FailToPass
1: 0.00 - Fail	1: 0.50 - Fail	1: 0.50 - Fail
2: 0.50 - Fail	2: 0.40 - Fail	2: 0.60 - Fail
3: 0.75 - Pass	3: 0.30 - Fail	3: 0.70 - Pass
4: 0.65 - Pass	4: 0.20 - Fail	4: 0.65 - Pass
5: 0.55 - Fail	5: 0.10 - Fail	5: 0.60 - Fail
6: 0.60 - Fail	6: 0.00 - Fail	
7: 0.65 - Pass		
Result: 0.65	Result: 0.00	Result: 0.65
Success	Stuck at Fail	Success

Algorithm = Binary

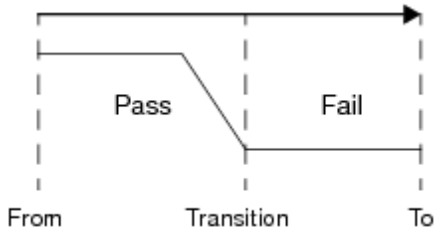
Transition = PassToFail	Transition = FailToPass
1: 0.50 - Fail	1: 0.50 - Fail
2: 0.20 - Fail	2: 0.75 - Pass
3: 0.05 - Fail	3: 0.60 - Fail
4: 0.00 - Fail	4: 0.65 - Pass
Result: 0.00	Result: 0.65
Stuck at Fail	Success

Algorithm = Linear

ResultsCheck = First

Transition = PassToFail	Transition = FailToPass
1: 0.00 - Fail	1: 0.00 - Fail
	2: 0.05 - Fail
	3: 0.10 - Fail
	4: 0.15 - Fail
	5: 0.20 - Fail
	6: 0.25 - Fail
	7: 0.30 - Fail
	8: 0.35 - Fail
	9: 0.40 - Fail
	10: 0.45 - Fail
	11: 0.50 - Fail
	12: 0.55 - Fail
	13: 0.60 - Fail
	14: 0.65 - Pass
Result: 0.00	Result: 0.65
Stuck at Fail	Success

Pass To Fail Transition at 0.61
From: 0.00, To: 1.00, Step: 0.05



Algorithm = Successive

Transition = First	Transition = PassToFail	Transition = FailToPass
--------------------	-------------------------	-------------------------

1: 0.00 - Pass	1: 0.50 - Pass	1: 0.50 - Pass
2: 0.50 - Pass	2: 0.75 - Fail	2: 0.25 - Pass
3: 0.75 - Fail	3: 0.65 - Fail	3: 0.00 - Pass
4: 0.65 - Fail	4: 0.55 - Pass	
5: 0.55 - Pass	5: 0.60 - Pass	
6: 0.60 - Pass	6: 0.65 - Fail	
7: 0.65 - Fail		
Result: 0.60	Result: 0.60	Result: 0.00
Success	Success	Stuck at Pass

Algorithm = Binary

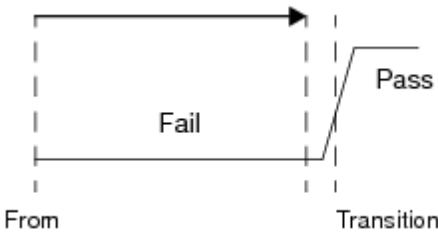
Transition = PassToFail	Transition = FailToPass
1: 0.50 - Pass	1: 0.50 - Pass
2: 0.75 - Fail	2: 0.20 - Pass
3: 0.60 - Pass	3: 0.05 - Pass
4: 0.65 - Fail	4: 0.00 - Pass
Result: 0.60	Result: 0.00
Success	Stuck at Pass

Algorithm = Linear
ResultsCheck = First

Transition = PassToFail	Transition = FailToPass
1: 0.00 - Pass	1: 0.00 - Pass
2: 0.05 - Pass	
3: 0.10 - Pass	
4: 0.15 - Pass	
5: 0.20 - Pass	
6: 0.25 - Pass	
7: 0.30 - Pass	
8: 0.35 - Pass	
9: 0.40 - Pass	
10: 0.45 - Pass	
11: 0.50 - Pass	

12: 0.55 - Pass	
13: 0.60 - Pass	
14: 0.65 - Fail	
Result: 0.60	Result: 0.00
Success	Stuck at Pass

Fail To Pass Transition at 1.1
From: 0.00, To: 1.00, Step: 0.05



Algorithm = Successive

Transition = First	Transition = PassToFail	Transition = FailToPass
1: 0.00 - Fail	1: 0.50 - Fail	1: 0.50 - Fail
2: 0.50 - Fail	2: 0.40 - Fail	2: 0.60 - Fail
3: 0.75 - Fail	3: 0.30 - Fail	3: 0.70 - Fail
4: 1.00 - Fail	4: 0.20 - Fail	4: 0.80 - Fail
	5: 0.10 - Fail	5: 0.90 - Fail
	6: 0.00 - Fail	6: 1.00 - Fail
Result: 1.00	Result: 0.00	Result: 1.00
Stuck at Fail	Stuck at Fail	Stuck at Fail

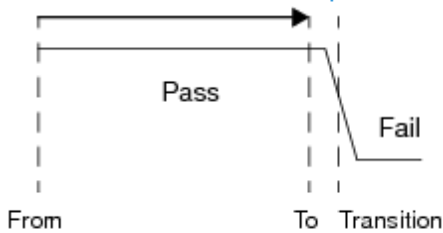
Algorithm = Binary

Transition = PassToFail	Transition = FailToPass
1: 0.50 - Fail	1: 0.50 - Fail
2: 0.20 - Fail	2: 0.75 - Fail
3: 0.05 - Fail	3: 0.90 - Fail
4: 0.00 - Fail	4: 0.95 - Fail
	5: 1.00 - Fail
Result: 0.00	Result: 0.00
Stuck at Fail	Stuck at Fail

Algorithm = Linear
ResultsCheck = First

Transition = PassToFail	Transition = FailToPass
1: 0.00 - Fail	1: 0.00 - Fail
	2: 0.05 - Fail
	3: 0.10 - Fail
	4: 0.15 - Fail
	5: 0.20 - Fail
	6: 0.25 - Fail
	7: 0.30 - Fail
	8: 0.35 - Fail
	9: 0.40 - Fail
	10: 0.45 - Fail
	11: 0.50 - Fail
	12: 0.55 - Fail
	13: 0.60 - Fail
	14: 0.65 - Fail
	15: 0.70 - Fail
	16: 0.75 - Fail
	17: 0.80 - Fail
	18: 0.85 - Fail
	19: 0.90 - Fail
	20: 0.95 - Fail
	21: 1.00 - Fail
Result: 0.00	Result: 1.00
Stuck at Fail	Stuck at Fail

Pass To Fail Transition at 1.1
From: 0.00, To: 1.00, Step: 0.05



Algorithm = Successive

Transition = First	Transition = PassToFail	Transition = FailToPass
1: 0.00 - Pass	1: 0.50 - Pass	1: 0.50 - Pass
2: 0.50 - Pass	2: 0.75 - Pass	2: 0.25 - Pass
3: 0.75 - Pass	3: 1.00 - Pass	3: 0.00 - Pass
4: 1.00 - Pass		
Result: 1.00	Result: 1.00	Result: 0.00
Stuck at Pass	Stuck at Pass	Stuck at Pass

Algorithm = Binary

Transition = PassToFail	Transition = FailToPass
1: 0.50 - Pass	1: 0.50 - Pass
2: 0.75 - Pass	2: 0.20 - Pass
3: 0.90 - Pass	3: 0.05 - Pass
4: 0.95 - Pass	4: 0.00 - Pass
5: 1.00 - Pass	
Result: 1.00	Result: 0.00
Stuck at Pass	Stuck at Pass

Algorithm = Linear

ResultsCheck = First

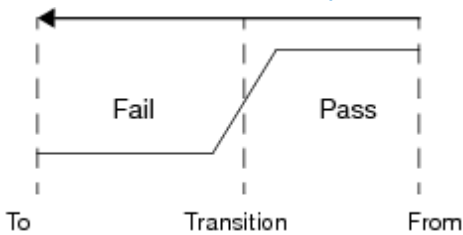
Transition = PassToFail	Transition = FailToPass
1: 0.00 - Pass	1: 0.00 - Pass
2: 0.05 - Pass	
3: 0.10 - Pass	
4: 0.15 - Pass	
5: 0.20 - Pass	
6: 0.25 - Pass	
7: 0.30 - Pass	
8: 0.35 - Pass	
9: 0.40 - Pass	
10: 0.45 - Pass	
11: 0.50 - Pass	
12: 0.55 - Pass	

13: 0.60 - Pass
14: 0.65 - Pass
15: 0.70 - Pass
16: 0.75 - Pass
17: 0.80 - Pass
18: 0.85 - Pass
19: 0.90 - Pass
20: 0.95 - Pass
21: 1.00 - Pass

Result: 1.00
Stuck at Pass

Result: 0.00
Stuck at Pass

Fail to Pass Transition at 0.61
From: 1.00, To: 0.00, Step: -0.05



Algorithm = Successive

Transition = First	Transition = PassToFail	Transition = FailToPass
1: 1.00 - Pass	1: 0.50 - Fail	1: 0.50 - Fail
2: 0.50 - Fail	2: 0.60 - Fail	2: 0.40 - Fail
3: 0.60 - Fail	3: 0.70 - Pass	3: 0.30 - Fail
4: 0.70 - Pass	4: 0.65 - Pass	4: 0.20 - Fail
5: 0.65 - Pass	5: 0.60 - Fail	5: 0.10 - Fail
6: 0.60 - Fail		6: 0.00 - Fail
Result: 0.65	Result: 0.65	Result: 0.00
Success	Success	Stuck at Fail

Algorithm = Binary

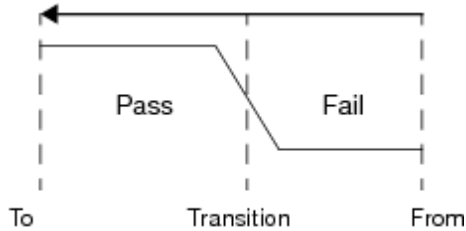
Transition = PassToFail	Transition = FailToPass
-------------------------	-------------------------

1: 0.50 - Fail	1: 0.50 - Fail
2: 0.80 - Pass	2: 0.25 - Fail
3: 0.65 - Pass	3: 0.10 - Fail
4: 0.60 - Fail	4: 0.05 - Fail
	5: 0.00 - Fail
Result: 0.65	Result: 1.00
Success	Stuck at Fail

Algorithm = Linear
ResultsCheck = First

Transition = PassToFail	Transition = FailToPass
1: 1.00 - Pass	1: 1.00 - Pass
2: 0.95 - Pass	
3: 0.90 - Pass	
4: 0.85 - Pass	
5: 0.80 - Pass	
6: 0.75 - Pass	
7: 0.70 - Pass	
8: 0.65 - Pass	
9: 0.60 - Fail	
Result: 0.65	Result: 1.00
Success	Stuck at Pass

Pass To Fail Transition at 0.61
From: 1.00, To: 0.00, Step: -0.05



Algorithm = Successive

Transition = First	Transition = PassToFail	Transition = FailToPass
--------------------	-------------------------	-------------------------

1: 1.00 - Fail	1: 0.50 - Pass	1: 0.50 - Pass
2: 0.50 - Pass	2: 0.75 - Fail	2: 0.25 - Pass
3: 0.60 - Pass	3: 0.65 - Fail	3: 0.00 - Pass
4: 0.70 - Fail	4: 0.55 - Pass	
5: 0.65 - Fail	5: 0.60 - Pass	
6: 0.60 - Pass	6: 0.65 - Fail	
Result: 0.60	Result: 0.60	Result: 0.00
Success	Success	Stuck at Pass

Algorithm = Binary

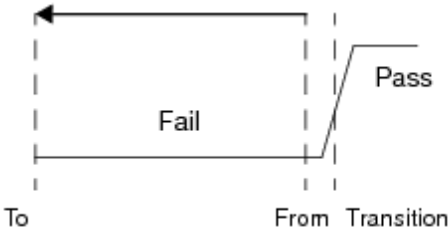
Transition = PassToFail	Transition = FailToPass
1: 0.50 - Pass	1: 0.50 - Pass
2: 0.25 - Pass	2: 0.80 - Fail
3: 0.10 - Pass	3: 0.65 - Fail
4: 0.05 - Pass	4: 0.60 - Pass
5: 0.00 - Pass	
Result: 0.00	Result: 0.60
Stuck at Pass	Success

Algorithm = Linear
ResultsCheck = First

Transition = PassToFail	Transition = FailToPass
1: 1.00 - Fail	1: 1.00 - Fail
	2: 0.95 - Fail
	3: 0.90 - Fail
	4: 0.85 - Fail
	5: 0.80 - Fail
	6: 0.75 - Fail
	7: 0.70 - Fail
	8: 0.65 - Fail
	9: 0.60 - Pass
Result: 1.00	Result: 0.60
Stuck at Fail	Success

Fail To Pass Transition at 1.1

From: 1.00, To: 0.00, Step: -0.05



Algorithm = Successive

Transition = First	Transition = PassToFail	Transition = FailToPass
1: 1.00 - Fail	1: 0.50 - Fail	1: 0.50 - Fail
2: 0.50 - Fail	2: 0.60 - Fail	2: 0.40 - Fail
3: 0.25 - Fail	3: 0.70 - Fail	3: 0.30 - Fail
4: 0.00 - Fail	4: 0.80 - Fail	4: 0.20 - Fail
	5: 0.90 - Fail	5: 0.10 - Fail
	6: 1.00 - Fail	6: 0.00 - Fail
Result: 0.00	Result: 1.00	Result: 0.00
Stuck at Fail	Stuck at Fail	Stuck at Fail

Algorithm = Binary

Transition = PassToFail	Transition = FailToPass
1: 0.50 - Fail	1: 0.50 - Fail
2: 0.80 - Fail	2: 0.25 - Fail
3: 0.95 - Fail	3: 0.10 - Fail
4: 1.00 - Fail	4: 0.05 - Fail
	5: 0.00 - Fail
Result: 1.00	Result: 1.00
Stuck at Fail	Stuck at Fail

Algorithm = Linear

ResultsCheck = First

Transition = PassToFail	Transition = FailToPass
-------------------------	-------------------------

1: 1.00 - Fail

1: 1.00 - Fail

2: 0.95 - Fail

3: 0.90 - Fail

4: 0.85 - Fail

5: 0.80 - Fail

6: 0.75 - Fail

7: 0.70 - Fail

8: 0.65 - Fail

9: 0.60 - Fail

10: 0.55 - Fail

11: 0.50 - Fail

12: 0.45 - Fail

13: 0.40 - Fail

14: 0.35 - Fail

15: 0.30 - Fail

16: 0.25 - Fail

17: 0.20 - Fail

18: 0.15 - Fail

19: 0.10 - Fail

20: 0.05 - Fail

21: 0.00 - Fail

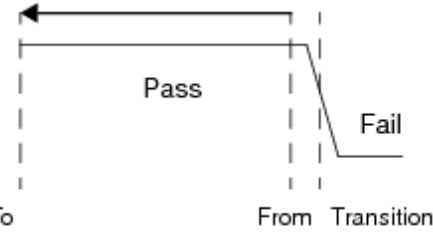
Result: 1.00

Stuck at Fail

Result: 0.00

Stuck at Fail

Pass To Fail Transition at 1.1
From: 1.00, To: 0.00, Step: -0.05



Algorithm = Successive

Transition = First

Transition = PassToFail

Transition = FailToPass

1: 1.00 - Pass	1: 0.50 - Pass	1: 0.50 - Pass
2: 0.50 - Pass	2: 0.25 - Pass	2: 0.75 - Pass
3: 0.25 - Pass	3: 0.00 - Pass	3: 1.00 - Pass
4: 0.00 - Pass		
Result: 0.00	Result: 0.00	Result: 1.00
Stuck at Pass	Stuck at Pass	Stuck at Pass

Algorithm = Binary

Transition = PassToFail	Transition = FailToPass
1: 0.50 - Pass	1: 0.50 - Pass
2: 0.25 - Pass	2: 0.80 - Pass
3: 0.10 - Pass	3: 0.95 - Pass
4: 0.05 - Pass	4: 1.00 - Pass
5: 0.00 - Pass	
Result: 0.00	Result: 1.00
Stuck at Pass	Stuck at Pass

Algorithm = Linear
ResultsCheck = First

Transition = PassToFail	Transition = FailToPass
1: 1.00 - Pass	1: 1.00 - Pass
2: 0.95 - Pass	
3: 0.90 - Pass	
4: 0.85 - Pass	
5: 0.80 - Pass	
6: 0.75 - Pass	
7: 0.70 - Pass	
8: 0.65 - Pass	
9: 0.60 - Pass	
10: 0.55 - Pass	
11: 0.50 - Pass	
12: 0.45 - Pass	
13: 0.40 - Pass	

14: 0.35 - Pass
15: 0.30 - Pass
16: 0.25 - Pass
17: 0.20 - Pass
18: 0.15 - Pass
19: 0.10 - Pass
20: 0.05 - Pass
21: 0.00 - Pass

Result: 0.00
Stuck at Pass

Result: 1.00
Stuck at Pass

	
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CZOtherFeatures.4.1

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Other Characterization Features

Other Characterization Features

This section describes characterization features other than those provided by Characterization Studio. The following topics are available:

- Quick Margin
- Running Margins on a Flow from VBT

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CZOtherFeatures.4.2

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Other Characterization Features : Quick Margin

Quick Margin

Instead of using Characterization Studio, you can also run a Margin on one or more specs from a Specs sheet in DataTool. This is called Quick Margin. The specs are margined for the entire flow. Two columns are appended to the Specs sheet indicating the positive and negative margins. You can use default values for the Quick Margin, or you can edit the setup in Characterization Tool, select the specs on the sheet, and run Quick Margin again.

Note the following:

- You can use Quick Margin only when a job is not running.
- A test program that includes a concurrent sequence cannot use Quick Margin. If you disable concurrency or force concurrent subflows to run serially, then Quick Margin will work properly. For more information, see [Creating Programs for Concurrent Test](#).

Setting Up Quick Margin

Go to the AC, DC, or Global Specs sheet. You can select one or more specs on the sheet. Right-click, and from the context menu select Quick Margin.

IG-XL looks for a characterization setup named `tl_DefaultSetup_QuickMargin<spectype>` in memory or on the Characterization sheet: The spectype can be AC, DC, or GL.

- If the setup exists, it is loaded and the newly selected specs are entered to be margined. This is useful when you want to use nondefault values, as described below.
- If the setup does not exist, it is created for the selected specs using default values for all parameters. These are shown below.

The system binds the setup to all active flow tables and executes the flow. For each spec, it uses Min and Max as the From/To range, if they are defined, Otherwise, it uses a range around the programmed value. IG-XL merges the result edges as a percentage for each spec, regardless of the range used for any given context.

To display the results, IG-XL adds two columns to the Specs sheet immediately after the Comment column, Pos Margin and Neg Margin. IG-XL populates these columns with the percentage value of each spec that shows the narrowest passing band for all tests that use the spec. These are not DataTool columns

Because any given spec could be operating in multiple ranges depending on the selector used in each context, the results must be merged as a percentage. So, regardless of whether the spec is margined as Min/Max or as a range around the current value, the edges are calculated as a percentage around the center of the range, and the reported percentage will be the worst case (narrowest band) of all instances measured.

Note the following:

- NaN (not a number) in a Characterization Studio result has no association with the percentage. Normally, a Margin result displays either P-> F or F->P for NaN values. In Quick Margin, however, NaN values are always displayed as NaN in both columns.

- In all cases, Quick Margin ignores values of NaN.

For example, consider a Margin in 10 steps (11 points):

Spec1 range in context1 = 100ns to 110ns (nom = 105ns)

Spec1 range in context2 = 120ns to 140ns (nom = 130ns)

Context1 - FFPPPPPPFFFF = Neg Marg % = 2.86, Pos Marg % = 1.9

Context2 - FFFFPPPPFFFF = Neg Marg % = 1.54, Pos Marg % = 4.61

The reported margin range is the worst case, so Neg Marg % = 1.54, and Pos Marg % = 1.9.

If the current context is not dependent on the spec being margined, the results are disregarded. A failing test could make it appear that a spec has no margin when in fact it does.

Quick Margin Default Values

The default values used for `tl_DefaultSetup_QuickMargin<spectype>` are as follows:

When creating a setup using the single click option for AC specs, IG-XL uses the following default values.

- TestMethod = Retest

- OutputFormat = Legacy

- [OutputDestinations](#) = none selected
- [Type](#) = as determined by the specs chosen
- [Name](#) = the spec(s) selected
- [Algorithm](#) = Linear
- [CalculatedField](#) = [StepSize](#)
- [From](#) = Min
- [To](#) = Max
- [Steps](#) = 20

To use nondefault values, use [Characterization Studio](#) to open the `tl_DefaultSetup_QuickMargin<spectype>` for the type of spec you want to characterize. Make your changes and save the setup to memory or to the Characterization sheet. Then go back to the Specs sheet, select the specs, and run Quick Margin again.

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CZOtherFeatures.4.3



Other Characterization Features : Running Margins on a Flow from VBT

Running Margins on a Flow from VBT

You can use VBT statements to get margin results for specified instances in a flow table. You create a margin setup that specifies the test instances. You then associate the setup with a Flow Table sheet or with a concurrent sequence step. When you execute the flow, the margin setup is executed for each test instance specified in the setup.

Creating the Setup

Open or create a Margin setup. On the General tab, under Margin Parameters, click the ellipsis button next to the Instances field:



Instances Parameter

Clicking the ellipsis button brings up the InstanceSelector. The Available window lists the test instances. You can use a wildcard filter to reduce the number of instances displayed. Move the instances you want to the Selected window and click OK.

Specify the parameters for the rest of the setup: parameters to vary, range, output options, and so on. Note that any Contexts or Patterns that you specify are ignored when the setup is run from the flow. They are used only if you run the setup from Characterization Studio.

Associating the Setup with the Flow

You use a VBT method to associate the Margin setup with the flow. The method itself is `.CharacterizationSetup.Add(setupname)`. The full statement depends on which flow is to be associated with the setup:

- To associate the setup with the entire flow, use:

TheExec.Flow.CharacterizationSetups.Add(setupname)

- To associate the setup with a specific Flow Table sheet (for example, with one of several subflows in the job), use:

TheExec.Flow.FlowTables.Item(flowtablename).CharacterizationSetups.Add(setupname)

- To associate the setup with a specific concurrent sequence steps, use:

TheExec.Flow.Steps.Item(indx).CharacterizationSetups.Add(setupname)
TheExec.Flow.FlowTables.Item(flowTableName).Steps.Item(indx).CharacterizationSetups.Add(setupname)

When the specified flow is executed, the Margin setup is executed for each test instance selected in the setup.

You can add the VBT statement to the OnProgramLoaded exec interpose function if you want the Margin executed every time the job is run.

For more information, see TheExec.Flow.CharacterizationSetups.

			
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CZStudio.3.01



Characterization Studio

Characterization Studio

Characterization Studio provides interactive access to characterization setups. You can create, edit, validate, and run characterization setups. Characterization Studio provides a full graphics display of characterization output.

The following topics are available:

- [Starting Characterization Studio](#)
- [Characterization Studio Window](#)
- [Open Setups Tree View](#)
- [Setups Editor](#)
- [Validating and Running Setups](#)
- [Output Display: General Points](#)
- [Using the General Tab](#)
- [Shmoo](#)
- [Shmoo Display](#)

- [Margin](#)
- [Adjust, Adjust From, and Measure](#)
- [General Notes on Mode-Specific Parameters](#)

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CZStudio.3.02



Characterization Studio : Starting Characterization Studio

Starting Characterization Studio

Bring up Characterization Studio in one of the following ways:

- From DataTool:



One-Click Launch Group

On the IG-XL Tools tab, in the One-Click Launch group, click Characterization Studio. The window displays the most recently executed setup, if any. If no setup has been executed, the window is empty. Use Open Setup to open a setup: see [Open Setups Tree View](#).

- On a Characterization sheet, select a name in the Setup Name column, right-click, and select Edit. The window displays the setup.
- On a Flow Table sheet, select an Opcode Parameter cell where the opcode is characterize, right-click, select Edit, and then select Characterization. The window displays the setup specified in as the parameter.

Only one instance of Characterization Studio is allowed at a time.



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CZStudio.3.03

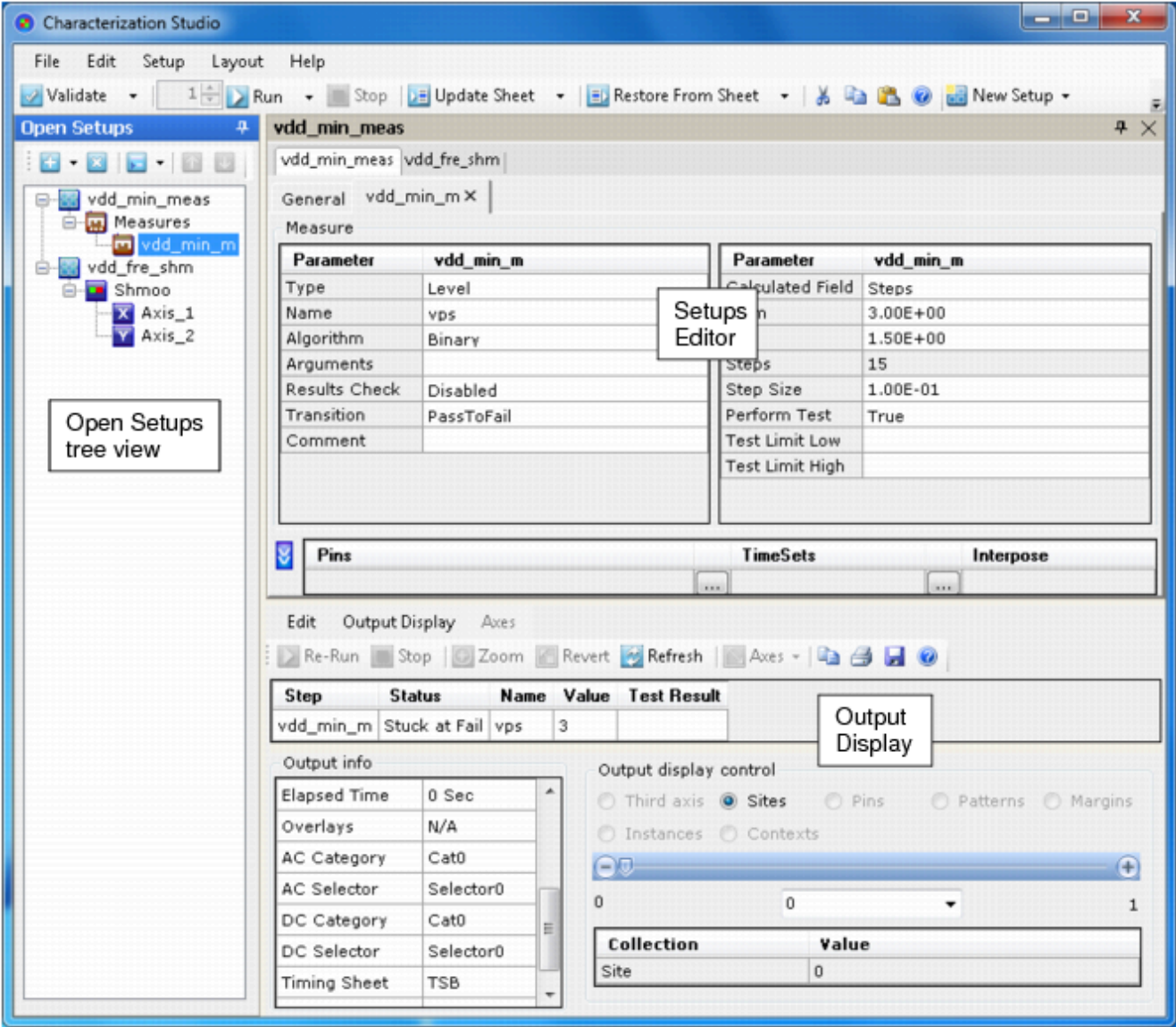
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Characterization Studio : Characterization Studio Window

Characterization Studio Window

The Characterization Studio window looks like this:



Characterization Studio Window

Changing and Restore the Window Layout

You can make the various windows of Characterization Studio float and dock at different positions. Grab the toolbar and pull it away from its location; controls appear that let you move the window up, down, right, or left by moving the window onto that control.

To restore the windows to their default position, use **Layout>Restore Default**.

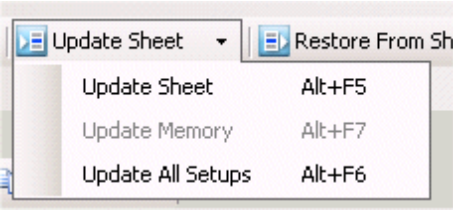
Split Buttons and Drop-Down Buttons

Some toolbar buttons are split buttons. For example:



Split Button

A split button combines two functions in a single control: executing the currently displayed command or selecting a different command. To execute the currently shown command, click the button icon or label. To select a different option, click the inverted black triangle at the far right and select the option you want; this executes the selection and makes it the new default command (during a session) until changed again:



Split Button with Options Displayed

A split button is similar in appearance to a drop-down button, in that both have the inverted black triangle. The difference is that, when you hover the pointer over the button, the split button has a black line separating the label from the triangle (see the **Update Sheet** button above).



A drop-down button does not have the black line.

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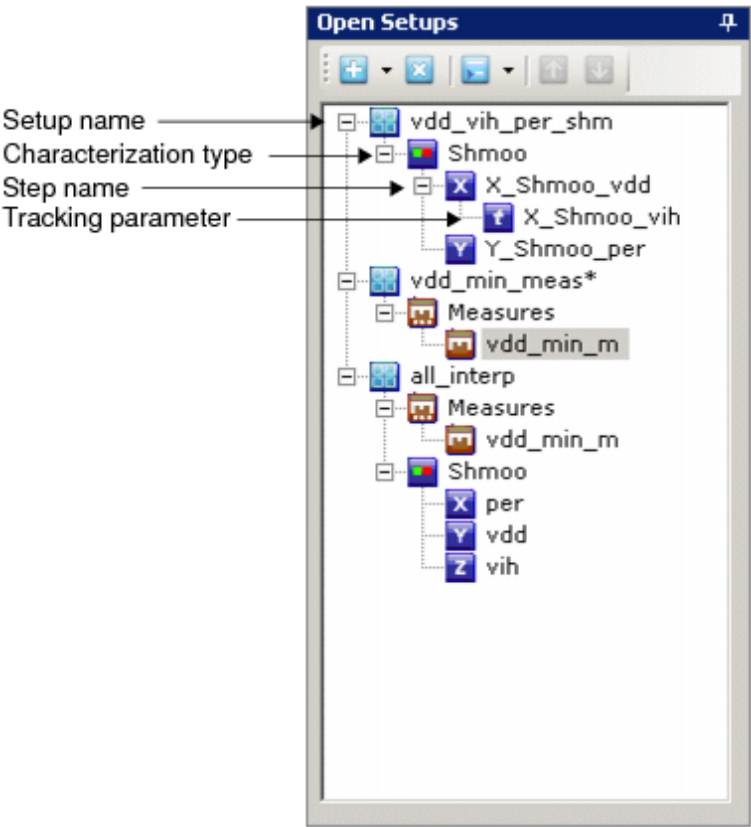
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Characterization Studio : Open Setups Tree View

Open Setups Tree View

The Open Setups tree view provides a graphical view of the characterization setups that you have opened for editing or execution. It lets you choose which setup and step you want displayed in the editor. You can also add new setups or new steps to existing setups, copy and paste setups and steps, and change the sequence of steps in a setup.



Open Setups Tree View

Each setup consists of one or more named steps. Each step defines one parameter that is to be varied. In the tree view, each setup is displayed as follows:

- | | |
|--|---|
| | <ul style="list-style-type: none">• The first, leftmost level is the setup name. |
| | <ul style="list-style-type: none">• Under the name, the steps are grouped by characterization type. The second level is the type name, such as Shmoo or Margin. If a setup has steps for different characterization types, there is one second-level entry for each type. |
| | <ul style="list-style-type: none">• The third level shows the step names of each type. For a shmoo, an icon indicates whether the step is the X, Y, or Z axis. If an axis has one or more tracking parameters, these are placed in a fourth level under the axis step and are marked with a “t” icon. |

If a setup name has an asterisk (*), this indicates that a value in the setup has been changed. The change can be at any level of the setup.

You can perform the following actions in the Open Setups tree view:

- | | |
|--|--|
| | <ul style="list-style-type: none">• Opening an Existing Setup |
| | <ul style="list-style-type: none">• Adding a New Setup |
| | <ul style="list-style-type: none">• Adding a Step to an Existing Setup |
| | <ul style="list-style-type: none">• Deleting a Step from a Setup |
| | <ul style="list-style-type: none">• Closing a Setup |
| | <ul style="list-style-type: none">• Copying and Inserting a Setup or Step |
| | <ul style="list-style-type: none">• Changing the Order of Steps and Setups |

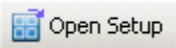
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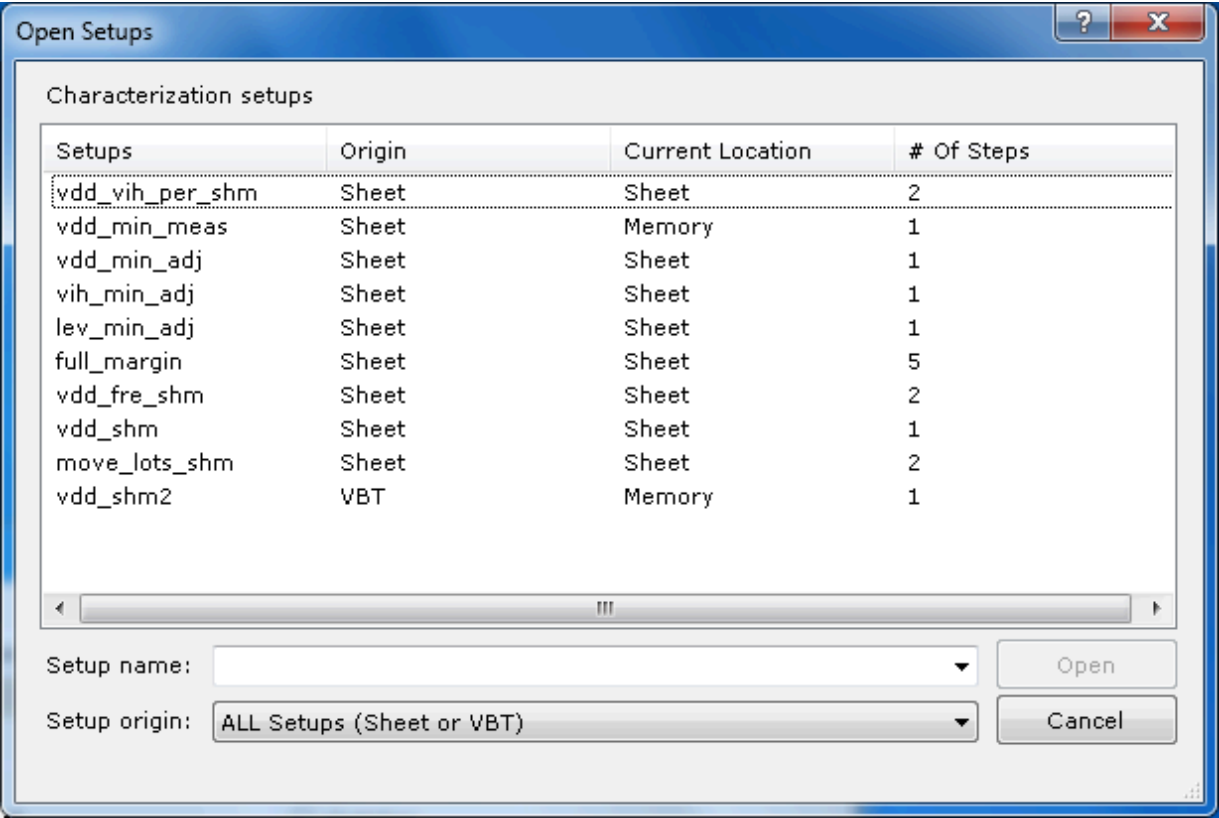
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Characterization Studio : Open Setups Tree View : Opening an Existing Setup

Opening an Existing Setup

 **Open Setup** To open an existing setup, click this button, or use File>Open Setup or Alt+O. The Open Setups window opens:



Open Setups Window

By default, the window displays all available setups, those defined on a Characterization sheet and those defined in VBT. You can use the Setup origin: control to display a subset of the available setups:

Sheet Setups	Display only those setups initially defined on a Characterization sheet.
VBT Setups	Display only those setups initially created by VBT during the current program execution.
ALL Setups (Sheet or VBT)	Display all setups. Default.

The window has these columns:

Setups	The setup name.
Origin	Where the setup was initially created, on a Characterization sheet or in VBT.
Current Location	Whether the setup is currently stored on the Sheet or in Memory. A setup created by VBT is located in memory. A setup defined on the sheet but modified by VBT is located in memory; in this case, opening the setup opens the modified version, that is, the version currently located in memory, not the original unmodified version as initially create on the sheet. For example, in the screenshot above, vdd_min_meas was created on the sheet but have been modified, so it is now located in memory. The setup vdd_shm2 was created in VBT and therefore is located in memory.
# of Steps	The number of steps in the setup.

You can sort on any column by clicking the column header.

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Characterization Studio : Open Setups Tree View : Adding a New Setup

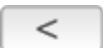
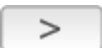
Adding a New Setup

 To create a new setup, click this button or use File>New Setup. A drop-down list lets you select the kind of setup to create: Adjust, Margin, Measure, or 1D, 2D, or 3D Shmoo. If you select one of these types, the new setup is created with the structure required for the type: that is, with the second-level characterization type and the third-level step (or steps for a 2D or 3D Shmoo). You can also create an Empty Setup, which will have only the first-level setup name.

When you select from the list, you are prompted for a setup name. You can change the setup name and any step name later: use Setup>Rename or F2. The setup name must be unique across all active Characterization sheets.

The new setup is created below any currently displayed setups.

Once you create a setup, you edit it to assign its parameters the desired values.

	
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Characterization Studio : Open Setups Tree View : Adding a Step to an Existing Setup

Adding a Step to an Existing Setup

You can add a step to an existing setup. Select the node in the tree view to which you want to add the step. This can be any of the three levels in the setup:

- The setup name
- The characterization type
- A step within the setup

Then use one of these methods to bring up the list of valid steps:

-  Click the inverted triangle on this button on the Open Setups tree view pane.
- Right-click the node in the tree view, select Add from the context menu, and display the pull-right menu.
- Use the Setup>Add command and display the pull-right menu.

The list contains all possible steps that could be added, but only those that are valid for the node you have selected are enabled. For example, take this setup:

MeasAndShmoo
 Measures
 vdd_min_M
 Shmoo

X - per
Y - vdd

With any node selected, the list will have enabled Measure, Adjust, Adjust From, and Z axis, because these are all valid for the setup as it exists and Characterization Studio knows where each would be inserted. So, for example, even if you have the Measure step selected and you say you want to add a Z axis, the tool knows that the Z axis is added after the Y axis in the Shmoo setup. On the other hand, to add a tracking parameter, you must select either the X or Y axis before that option is enabled, because that is the only way the Studio can know which axis is to have the tracking parameter. If the Shmoo had just one axis, the tracking parameter option would always be enabled, because the Studio would know that it was to be added to the X axis.

You can also use an Alt+ key sequence to add each kind of step. Select a node in a setup and type the sequence. If adding the step is valid, it is added; otherwise, it is ignored. Alt+ key sequences are:

Key Sequences for Adding Steps

Step	Key Sequence
Shmoo	Alt+H
Axis	Alt+X
Tracking Parameter	Alt+T
Adjust	Alt+J
Adjust From	Alt+F
Measure	Alt+M
Margin	Alt+R

When you add a step, it has a default name. To rename it, use Setup>Rename or F2.

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Characterization Studio : Open Setups Tree View : Deleting a Step from a Setup

Deleting a Step from a Setup

You can delete a step from a setup. Note that you cannot delete an entire setup using Characterization Studio.

To delete a step, select the step in the tree view. You can select a specific single step. You can also select a second-level characterization type item to delete all steps of that type. For example, take this setup:

MeasMinMax

Measures

vdd_min_M

vdd_max_M

You can select either of the Measures to delete that one step. If you select the "Measures" node, you delete both measures.

Once you select the node to be deleted, do one of these to delete it:

-  Click the Delete button on the Open Setups toolbar.
- Use the Setup>Delete command.
- Right-click and select Delete from the context menu.
- Press the Delete key.

Note that, for a Shmoo, the first step or axis is always X, the second Y, and the third Z. If you delete the X axis, the previous Y axis becomes the new X axis, and the previous Z axis becomes the new Y axis.

If you copy a Shmoo axis that has tracking parameters, the tracking parameters are copied as well.

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Characterization Studio : Open Setups Tree View : Closing a Setup

Closing a Setup

Closing a setup removes it from the Open Setups tree view. To close a setup, select it in the tree view, and do one of these:

- Use the File>Close Setup command or the Alt+C sequence. You can have any item in the setup selected for these to work.
- Right-click and select Close from the context menu. You must have the setup name selected for Close to appear on the context menu.

If the setup has been modified, you are asked if you want to save the changes before closing the setup. If the setup was initially defined on a Characterization sheet, you are asked if you want to update the sheet. If it was defined in VBT, you are asked if you want to update memory. If the setup was defined on a sheet but modified in VBT, you are asked if you want to update both.

	
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Characterization Studio : Open Setups Tree View : Copying and Inserting a Setup or Step

Copying and Inserting a Setup or Step

You can copy an entire setup within the tree view to use as the basis for a new setup. You can copy a step from one setup and insert it into the same setup or a different setup. You can also copy the second-level characterization type, with all of its steps, and insert it at a different location in the tree view.

To copy a setup or step, select the node in the Open Setups tree view. The node can be a top-level setup name, a second-level characterization type (for example, Measures or Shmoo), a third-level step or axis, or, for a Shmoo, a tracking parameter. Once you select the node, use one of these methods to copy it:

-  Press the Copy/Insert split button and select Copy Selected. You can also just click the button, because Copy Selected is the default.
- Right-click the node and select Copy/Insert>Copy Selected on the context menu.
- Use the Setup>Copy/Insert>Copy Selected command.
- Use Ctrl+C.

Note that this command is called Copy Selected, to distinguish it from the standard Copy command, which is available on Edit menu and on the Studio toolbar:

- Copy copies the value of a single field in the Editor for pasting into a different field.

- [Copy Selected](#) copies a setup or step in the Open Setups tree view.
- [Ctrl+C](#) is the shortcut for both. The proper command is executed, depending on context.

Once you copy the selected setup or step, select the location in the tree view where you want it inserted. You now have the following options, each with the indicated shortcut:

- [Insert Above](#) - [Atl+A](#)
- [Insert Into](#) - [Alt+I](#)
- [Insert Below](#) - [At+B](#)

You can copy any setup or step in the tree view. When you select a different location as the target, only insertions that are valid for the copied item at that location are enabled; if no insertion is valid at the target location, no insertion commands are enabled.

To perform the insertion, use one of these methods:

-  [Press the](#) Copy/Insert split button and select one of the enabled Insert commands.
- [Right-click the item and select one of the enabled](#) Insert commands on the context menu.
- [Use one of the](#) Setup>Copy/Insert>Insert commands.
- [Use one of the](#) Alt+ sequences shown above. If the sequence you use is not a valid insertion, nothing happens.

Note these points about insertions:

- For a Shmoo, the first step or axis is always X, the second Y, and the third Z. If you insert a new axis before the X axis, the inserted axis becomes the new X axis, the previous X axis becomes the new Y axis, and the previous Y axis becomes the new Z axis. If the Shmoo already has three axes, insertion commands are disabled if you have copied an axis.
- If you copy a Shmoo axis that has tracking parameters, the tracking parameters are copied as well.

- If the target is within the same setup, “_n” is appended to the name, where n is 1 for the first copy, 2 for the second copy, and so on. If the target is another setup that does not contain an object with the same name, no additional characters are appended to the name.

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Characterization Studio : Open Setups Tree View : Changing the Order of Steps and Setups

Changing the Order of Steps and Setups

You can change the order of execution for steps in a setup by changing the position of the step in the setup.

To move a step, select it in the tree view. If it is valid to move the step, the valid move commands and controls are enabled. You can move the third-level step nodes within a setup, or the fourth-level tracking parameters within a Shmoo axis. Note that the axes of a Shmoo are defined by their relative position: the first axis is the X axis, the second the Y, and the third the Z. If you move the current Y axis up one position, it becomes the X axis and is labelled as such.

To move the selected step, use one of these methods:

-  Use these buttons on the Open Setups toolbar.
- Use Setup>Move Up and Setup>Move Down.
- Right-click the setup in the tree view and, from the context menu, select Move Up and Move Down.
- Use Alt+Up and Alt+Down.
- Drag and drop the step within the setup.

You can also use these same commands and controls to move an entire setup within the tree view. This operation is simply a convenience during a debug session. You cannot save the new positions of the setups to the Characterization sheet.

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Characterization Studio : Setups Editor

Setups Editor

The editor is a tab-based interface where you can display and edit characterization setups shown in the Open Setups tree view. For new setups, the editor serves as a “fill in the blanks” set of forms for entering the settings for the setup. For existing setups, you can modify the settings and execute the setups without saving the modifications until you are ready to do so.

The editor and the Open Setups tree view are synchronized so that when you select a node in the tree view, the node’s tab is automatically selected in the editor; when you select a tab in the editor, the node is automatically selected in the tree view.

This section includes the following topics

- [Displaying Tabs and Buttons for Setup Settings](#)
- [Specifying Pins, TimeSets, and Interpose Functions](#)
- [Updating, Restoring, and Refreshing Setups](#)

The following sections cover the specific setting for different characterization modes:

- [Using the General Tab](#)
- [Shmoo](#)
- [Margin](#)

• Adjust, Adjust From, and Measure
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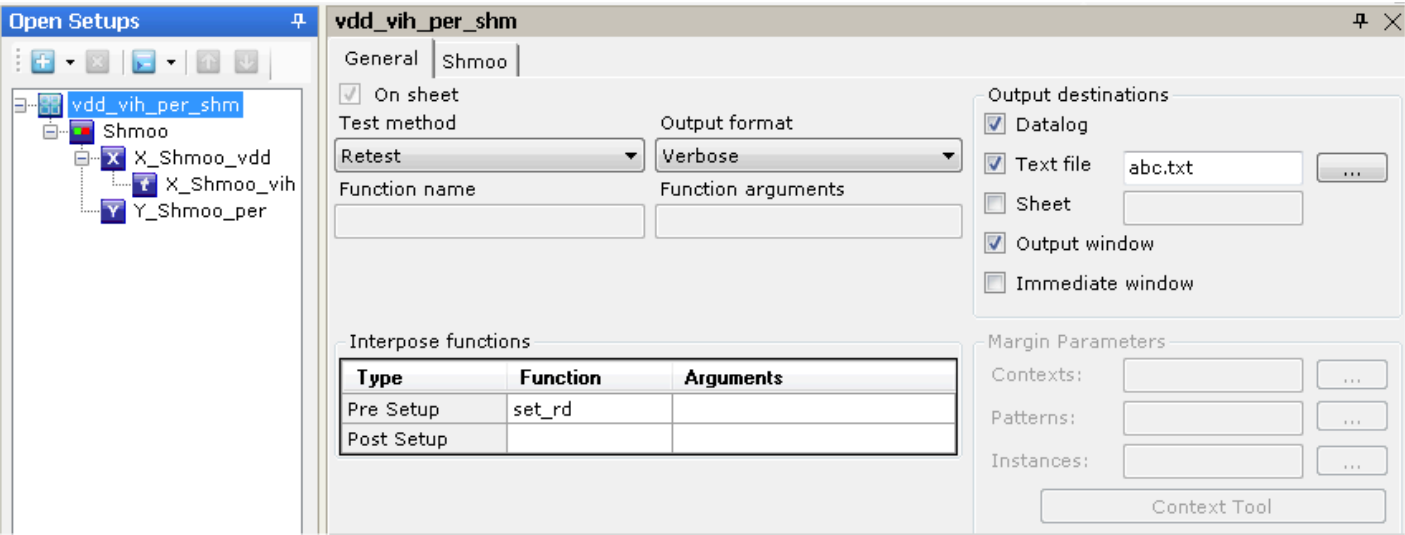
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Characterization Studio : Setups Editor : Displaying Tabs and Buttons for Setup Settings

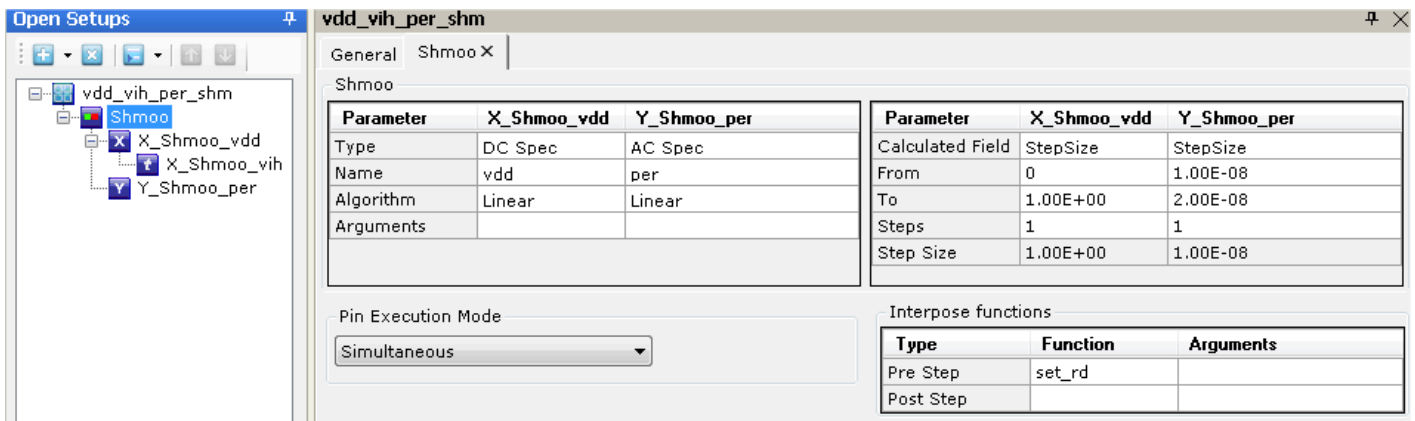
Displaying Tabs and Buttons for Setup Settings

The editor uses a set of tabs and buttons to control the display of characterization setups. When you click a node in the Open Setups tree view, a tab is added to the editor. By default, when you open or create a setup, the top-level setup node is selected. The corresponding editor tab is called General. The General tab contains settings that display to the entire setup:



Setups Editor General Tab

For details on editing these fields, see Using the General Tab. If you select the second-level characterization mode node (for example, Shmoo or Measures), the editor opens a tab with the name of the characterization mode:

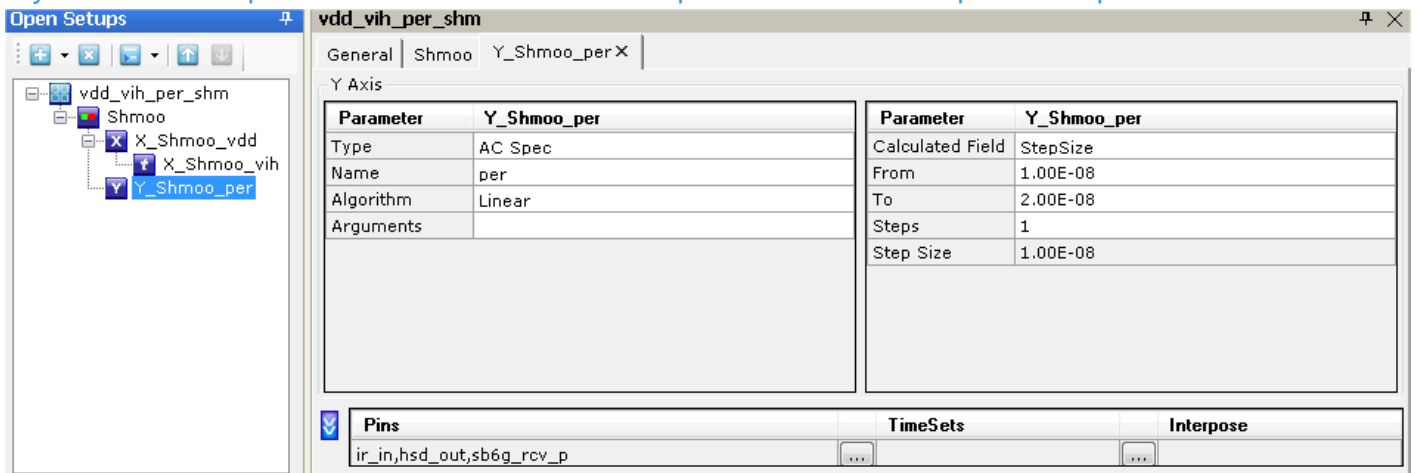


Setups Editor Characterization Mode Tab

This tab shows the settings for all steps of that characterization mode, or all axes of a Shmoo, as well as a few settings that apply to all steps of the mode.

There is a separate section on editing the tabs for each characterization mode.

If you select a step or a Shmoo axis, the editor opens a tab for that specific step or axis:



Setups Editor Characterization Step Tab

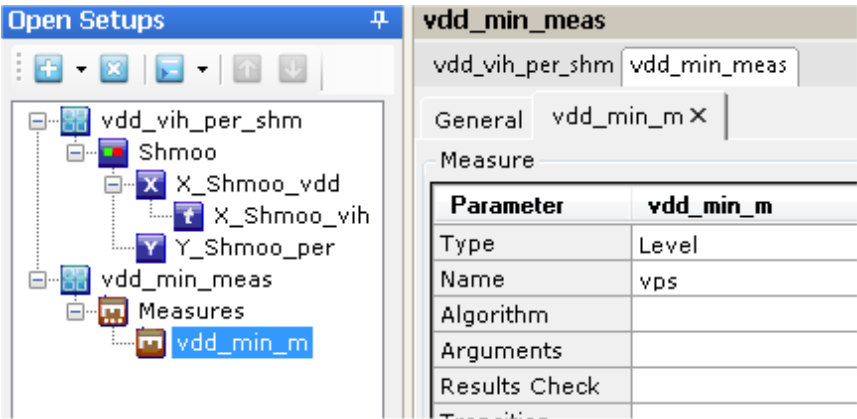
This tab shows the settings for this specific step or axis. Because pins, timesets, and interpose functions have step-specific settings, this tab lets you expose special editors for these items. See [Specifying Pins, TimeSets, and Interpose Functions](#).

If the step is a Shmoo axis that has a tracking parameter, settings for the tracking parameter are displayed also with those for the axis. You can also select the tracking parameter in the tree view to display a tab only for the tracking parameter.

Tabs are in the order in which you select items in the tree view. You can drag the tabs to change their order.

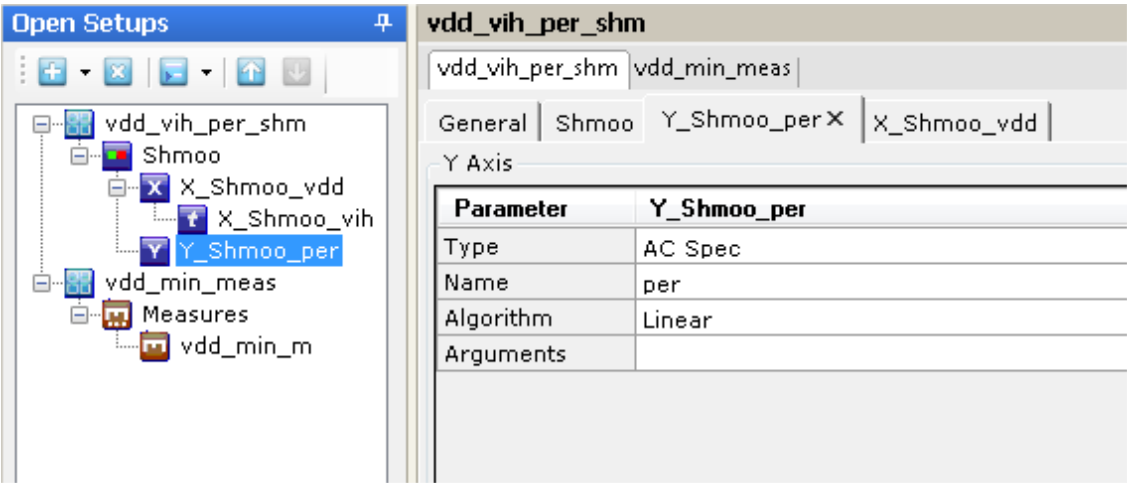
You can close a tab by clicking the x that appears when the tab is selected. (You cannot close a General tab.) Closing a tab does not discard its data and any modifications you have made. If you reopen the tab later, it displays the same data it had when you closed it.

When multiple setups are open, their names appear as buttons above the tabs:



Tabs with Multiple Setups

If you click a different button, the currently opened tabs for that setup are displayed:



Currently Opened Tabs

If only one setup is open, the button is not displayed.

You can display multiple setups at the same time using floating windows:

1. Grab the button for the setup and drag the setup window away from its location. Controls appear that let you select the location you want.
2. Drag the window onto the control for the direction you want relative to the rest of the setups.

You can restore the default layout using Layout>Restore Default or Ctrl+Shift+D.

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Characterization Studio : Setups Editor : Specifying Pins, TimeSets, and Interpose Functions

Specifying Pins, TimeSets, and Interpose Functions

At the level of a step, axis, or tracking parameter, you can specify a list of the pins and timesets whose parameter you want varied. If you do not specify a list, the default is to vary the parameter for all pins or for all timesets. In addition, you can also specify interpose functions to be called at the step level.

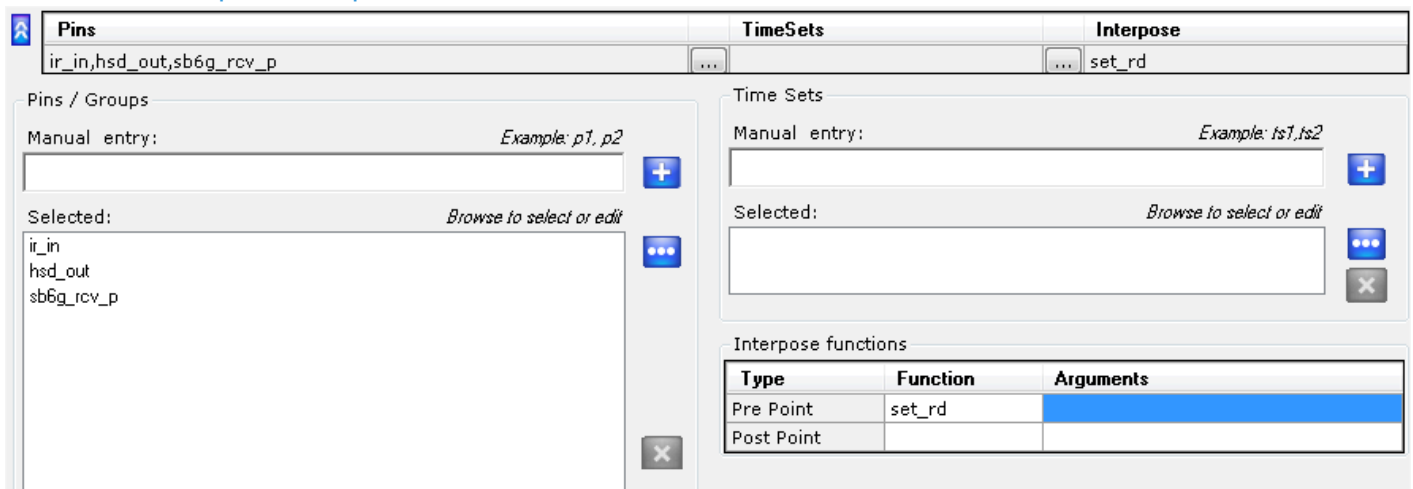
This section describes how to use the controls. For information on what the available values mean for each of these fields, see Pins, TimeSets, and Interpose.

When the tab for a step, axis, or tracking parameter is selected, you can expand and hide the panel containing the controls to help in specifying pins, timesets, and interpose functions:



Closed Controls Panel

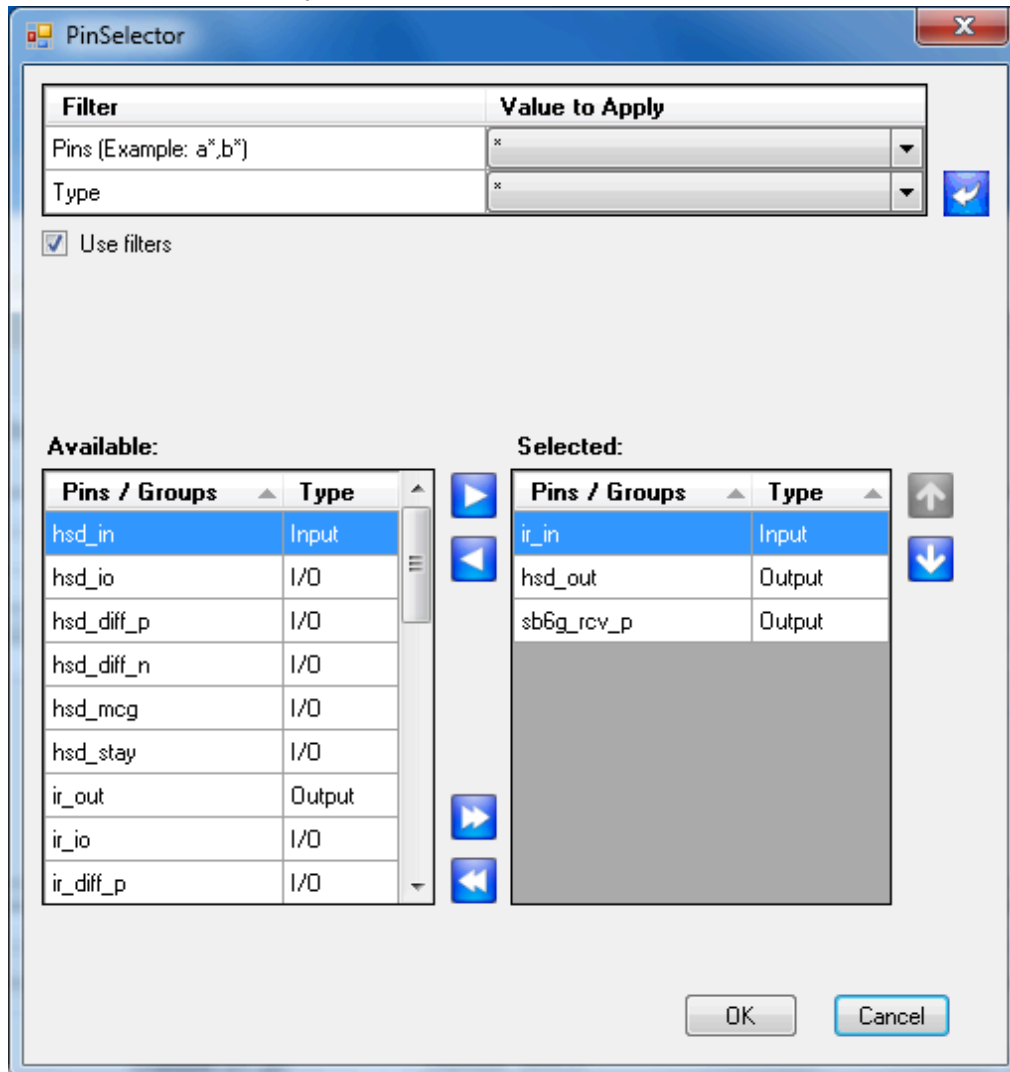
In its closed state, the panel displays a summary of any pins, timesets, and interpose functions currently specified for this step. In this state, you can click the ellipsis buttons (...) to bring up selectors for pins and timesets: these selectors are described later. You can also click the blue arrows at the left to expand the panel:



Open Controls Panel

For both Pins and TimeSets, you can use the Manual Entry: box to type in pins and pin groups and timesets. Items can be separated by a comma or space. Once you type in the items, click the + (plus sign) to add them to the Selected: window. They also appear in the Pins and TimeSets summary at the top of the panel.

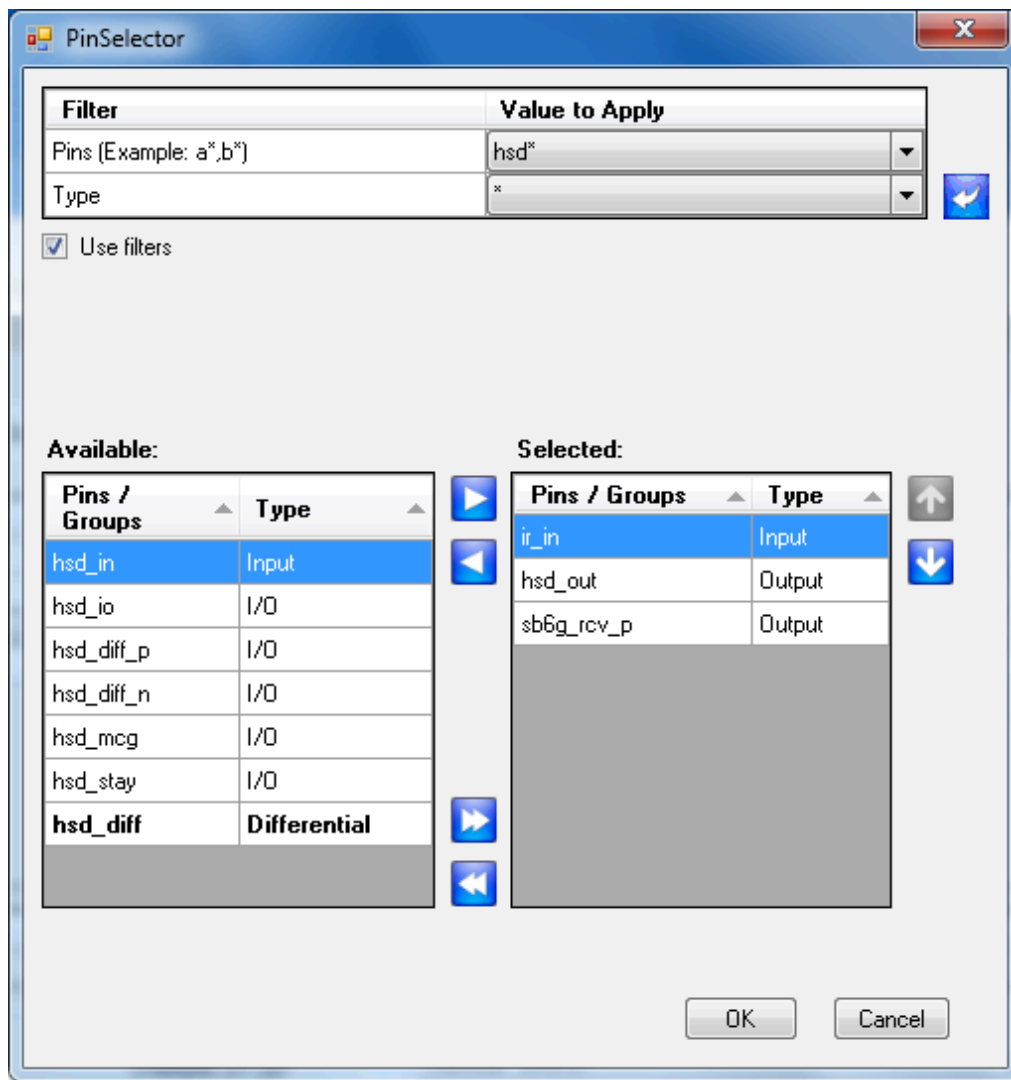
You can also use a Browser to select pins or timesets. Click the ellipsis button (...) next to the Selected: window; you can also use the ellipsis button in the summary. For Pins / Groups, the PinsSelector comes up:



PinsSelector

Select pins and pin groups under Available: and use the single right-arrow to move them to the Selected: window. Use the double right-arrow to move all pins. Use the left-arrows to deselect selected pins or all pins by moving them back from Selected: to Available:. In the windows, pins are represented in regular font and pin groups in bold font.

You can use filters to reduce the number of pins and groups visible under Available:



PinsSelector Using Filters

First check the Use filters checkbox. This lets you specify filters. In the Pins filter, you can enter a template using the standard wildcard characters * and ?. The Type filter is a drop-down list of all the pin types. You can use one or both filters.



Once you specify the filters, click this button to apply them. The filters pins and pin groups appear in the Available: list

The TimeSetsSelector window is similar to the PinsSelector window, except that it permits filtering only by name, using a wildcard template.

In addition to pins and timesets, the exposed panel also lets you specify the interpose functions and their arguments that are appropriate to this characterization mode. Enter the function names and arguments in the appropriate boxes.



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Characterization Studio : Setups Editor : Updating, Restoring, and Refreshing Setups

Updating, Restoring, and Refreshing Setups

When you make changes to setups in the editor, the changes are not immediately written to the Characterization sheet or to memory. Conversely, if changes are made to the setups directly on a sheet or through programming statements in memory, these are not immediately reflected in the editor. To align the editor values and those on a sheet or in memory, you have these options:

Update	You can update setups on a Characterization sheet or in memory with the changes you have made in the editor. Note that this is a separate step from permanently saving the changes in the workbook. The sheet is updated, but the changes are not permanently saved until you explicitly save the workbook.
Restore/Refresh	You can restore setups in the editor to the values they have on a Characterization sheet or refresh them using the values in memory. If any changes have been made to the setup on the sheet or in memory since the setup was opened or last updated, this causes them to be reflected in the editor. If no changes have been made, the setup is restored to the state it was in at the previous update or, if it has not been updated, at the time it was opened.

You can update or restore only the setup currently selected in the tree view. You can also update or restore all open setups.

The Update and Restore buttons on the Characterization Studio toolbar are split buttons. See [Split Buttons and Drop-Down Buttons](#).

To update, use one of these methods:

- [The Update](#) split button has these options:

- [Update Sheet](#): Update the sheet with the currently selected setup.

- [Update Memory](#): Update memory with the currently selected setup.

- [Update All Setups](#): Update all currently open setups. If a setup was originally defined on a Characterization sheet, the sheet is updated; if it was defined in VBT, memory is updated.

- [Use the File>Update](#) command, and select To Sheet, To Memory, or All.

- [Use a key sequence](#): Alt+F5 (Sheet), Alt+F7 (Memory), Alt+F8 (All).

When you try to update the sheet with a setup, the setup is first validated. Note these points:

- [If the setup is invalid, you are asked if confirm that you want to proceed with the update.](#)

- [If you update the sheet with a properly validated setup, the job does not need to be revalidated.](#)

For a new or copied setup, if the current job has multiple [Characterization sheets](#), Update Sheet or Update All Setups brings up a window with a drop-down list of all the Characterization sheets. Select the Characterization sheet where you want to store the setup.

[To restore or refresh, use one of these methods:](#)

- [The Restore](#) split button has these options:

- [Restore From Sheet](#): Restore the currently selected setup from the sheet.

- [Refresh From Memory](#): Refresh the currently selected setup from memory.

- [Restore/Refresh All](#): Restore all currently open setups. If a setup was originally defined on a Characterization sheet, it is restored from the sheet; if it was defined in VBT, it is refreshed from memory.

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Characterization Studio : Validating and Running Setups

Validating and Running Setups

You can validate and run characterization setups from Characterization Studio.

You can validate either the currently selected setups or all open setups using one of these methods:

-  Use the split button on the toolbar to select either:
 - [Validate](#) for the selected setup
 - [Validate all setups](#) for all open setups
- Use the [command](#) Setup>Validate>Validate Selected or Validate All.
- Use the [key sequence](#) Alt+V (validate selected) or Alt+L (validate all).

If validation succeeds, a message box indicates success. If validation fails, a message box indicates failure. Validation errors are written to the Test Program Output window.



To run a setup, select the setup to be run in the tree view. Then use the Run button. Options on the Run button are Run and Run No PreBody. By default, a reburst performs preconditioning. It runs the test PreBody and Body and then rebursts the pattern. To skip preconditioning, use Run No PreBody. This skips all preconditioning and simply calls reburst in a loop.

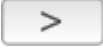
The number next to the Run button is enabled only for a Shmoo. It lets you accumulate results to create a composite Shmoo. See [Accumulating a Composite Shmoo](#).

The keyboard equivalent for Run is F5 and for Run No Prebody is F6.

If the setup you want to run has been modified but not validated, it is automatically validated before it is executed.

While the characterization is running, the Stop button is enabled.

Any run-time errors are displayed in the Test Program Output window.

		
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Characterization Studio : Output Display: General Points

Output Display: General Points

The Output Display presents the results of characterization runs. It is also a tab based interface and the results for multiple runs can be persisted, viewed, and compared. The Output Display appears differently depending on the types of steps in the setup. For example, Shmoo, Margin, and Measure have different requirements for displaying result data. Each Output Display also provides the ability to re-execute a setup or a subset of a setup, change the way the data is presented, or in some cases to explore characterization results in more detail.

The following topics are available:

- [Enabling Output Display](#)
- [Opening Stored Results From Memory](#)
- [Output Display Area: Overview](#)
- [Output Display Control](#)
- [Display Commands and Toolbar Buttons](#)
- [Merging Displayed Data](#)

For specific displays, see:

- [Shmoo Display](#)

•	Margin Display
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•	Adjust and Measure Display
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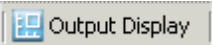
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Characterization Studio : Output Display: General Points : Enabling Output Display

Enabling Output Display

 By default, when you run a setup, its output is displayed, and, if applicable, the most optimized way to display the output is selected. The Output Display button lets you select nondefault behavior. Options depend on the characterization mode of the setup. For a Shmoo, you can disable display or choose the method of updating the display (which implies that display is enabled):

Disable	Do not create output displays.
Automatic	System optimizes output display method.
By point	Update display point-by-point.
By row	Update display row-by-row.
By Display	Update display only when done.

For a Margin, the button has these options:

Disable	Do not create output displays.
Automatic	System optimizes output display method

For a Measure or Adjust, the options are simply Disable or Enable.
The current Output Display setting applies only to the currently selected setup.
If you disable the output display, you can still view the display later by opening the results from memory. See Opening Stored Results From Memory.

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Characterization Studio : Output Display: General Points : Opening Stored Results From Memory

Opening Stored Results From Memory

When you run a setup, Characterization Studio stores the results in memory. You can later open and display the stored results by selecting the setup and using Open Results From Memory. This command is available from the Setup menu and from the right-click context menu when you have selected the setup in the tree view. The command displays the results of the most recent execution of the selected setup.

Even if you have disabled the output display, the results are stored and you can display them later using this command.

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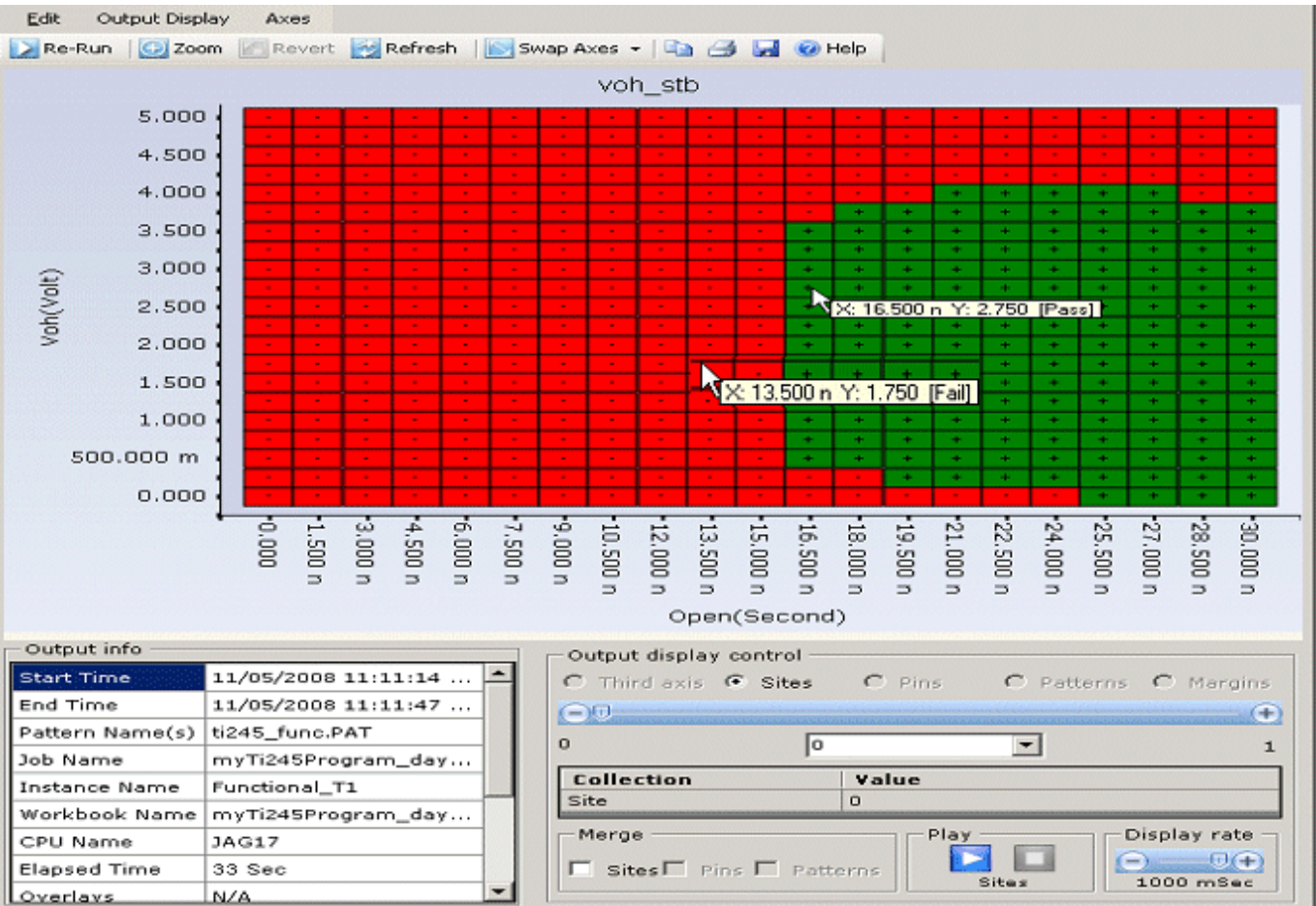
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Characterization Studio : Output Display: General Points : Output Display Area: Overview

Output Display Area: Overview

The output display area displays the results of executing a setup. The contents depend on the kind of setup executed, as do the kinds of operations that can be performed on the display.



Output Display Area

The output display area includes:

Command menus and toolbar	Provides options for rerunning a setup or modifying the display. Only commands that are valid for the current type of display are enabled. These commands are discussed in the sections on the different types of display.
Chart area	This area contains a graphic display of the characterization results except in the case of Measure or Adjust in which the data is displayed in a grid. The setup info provides details about the setup that are determined at run time. The display controls provide the ability to modify the view into the data.
Output info grid	This grid provides execution information: start and end times and elapsed execution time; pattern, job, instance, workbook, and CPU names; and context information.
Output display control	These controls are for viewing data in more than two dimensions, for example, for a multisite result, a pin-by-pin result, or a 3D Shmoo. See Output Display Control.
Context menu	Right-clicking the output display brings up a context menu that provides commands for rerunning or modifying the display. Some commands duplicate those on the menus and toolbar. Others are available only on this menu. The contents of the menu depend on the type of display and are discussed in separate sections.

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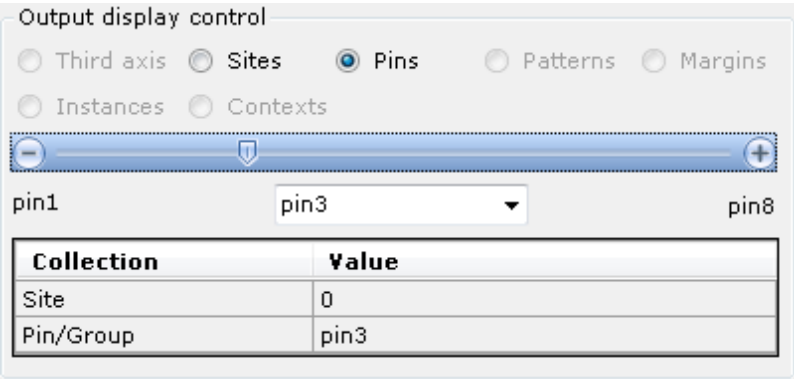
Characterization Studio : [Output Display: General Points](#) : Output Display Control

Output Display Control

The Output display control panel lets you display outputs of more than two dimensions, including:

- [A multipin, multipattern, or multisite display](#)
- [A Margin setup with multiple Margins, patterns, contexts, and others](#)
- [A three-dimensional Shmoo](#)

The panel has controls to let you navigate the third dimension.
Controls appearing on the panel depend on the characterization mode of the display. The simplest version, for Measure and Adjust, looks like this:



[Output Display Control Panel for Measure and Adjust](#)
The panels for all modes have these basic controls.

- [The radio buttons let you select the third dimension you want to navigate. Buttons are enabled for any dimension that has at least two items: for example, if there is only one pattern, Pattern](#)

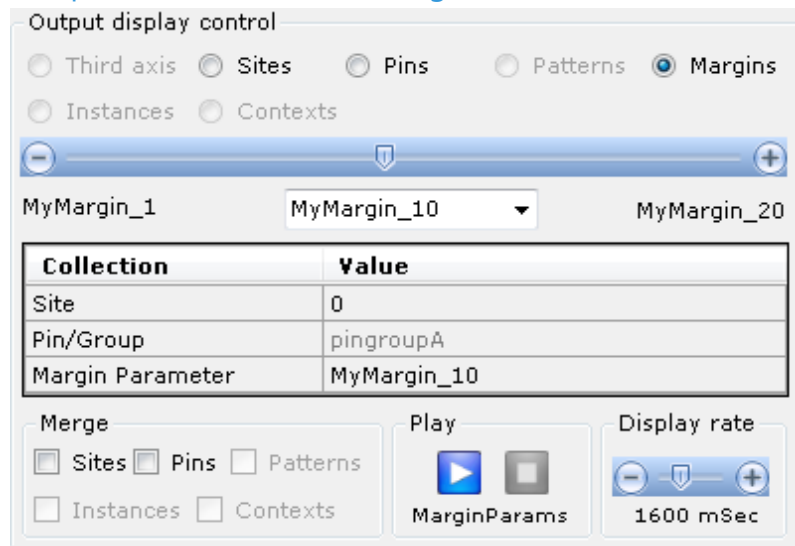
is not enabled. This example has multiple sites and multiple pins. You can select either as the third dimension.

- [The read-only table under the controls with headings labeled Collection and Value](#) shows the possible dimensions and their current settings.

The other controls let you navigate the third-dimension items. The two boxes on either side of the drop-down control show the range of values for the dimension. To select the item to display, use one of these methods:

- [Use the slider bar. Grab the slider and move it, or click the desired spot on the bar. Clicking the - moves the slider to the first item; clicking the + moves it to the last item.](#)
- [Select the item from the drop-down list.](#)

The panel for Shmoo and Margin has additional controls:



Output Display Control Panel for Shmoo and Margin

The Merge control lets you merge the selected dimensions. See [Merging Displayed Data](#).

The Play control lets you show all values for the selected dimension as a slide show. The Display rate slider specifies the time to display each value. Use the standard Play button to start the slide show and the standard Stop button to stop it.

[For a margin setup with multiple margin steps, you can select the Margins radio button.](#) The Play button then displays the margin steps one by one.

[On using the control to navigate a 3-D Shmoo, see Viewing the Third Axis.](#)

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Characterization Studio : Output Display: General Points : Display Commands and Toolbar Buttons

Display Commands and Toolbar Buttons

The display area has toolbar buttons, menu commands, and a right-click context menu commands that let you control the output display. This section lists and describes operations that are available. Most of these buttons and commands are available only for a Shmoo display. They are described in the section on Shmoo display: Shmoo Display Commands and Buttons. The ones that apply to all displays are described here.

Toolbar

The buttons on the display area toolbar, from left to right, are: Re-Run, Stop, Zoom, Revert, Refresh, and Axes. Only Refresh, described below, applies to non-Shmoo displays. The standard Copy, Print, and Save buttons are also available.

Menu Bar

The menus on the display area menu bar, from left to right, are:

Edit	Standard Cut, Copy, and Paste commands.
Output Display	Re-Run, Zoom, Revert, and Refresh. Only Refresh, described below, applies to non-Shmoo displays.
Axes	Enabled only for Shmoo.

Context Menu

When you right-click in the output display, a context menu becomes available. with these options, from top to bottom (note that some commands on the context menu have names different from those on the menu bar). The commands enabled for non-Shmoo displays are Launch Pattern Tool and Show/Hide Legend, described below.

Commands and Buttons

The following table describes the commands and buttons available:

Display Commands and Buttons

Command	Description
Axes	Shmoo only.
Launch PatternTool	Context menu only. Brings up PatternTool with the test's pattern open.
Show Detailed Point Info	Shmoo only.
Refresh	 Refresh Output Display>Refresh, Ctrl+F Refreshes the display with current data from memory.
Re-Run	Shmoo only.
Revert	Shmoo only.
Run Point and Restore	Shmoo only.
Show/Hide Legend	Context menu only. Show or hide the explanation of the colors of the boxes in the display.
Stop	Shmoo only.
Zoom	Shmoo only.

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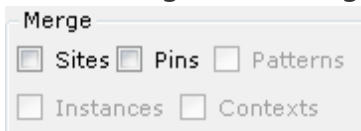
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Characterization Studio : Output Display: General Points : Merging Displayed Data

Merging Displayed Data

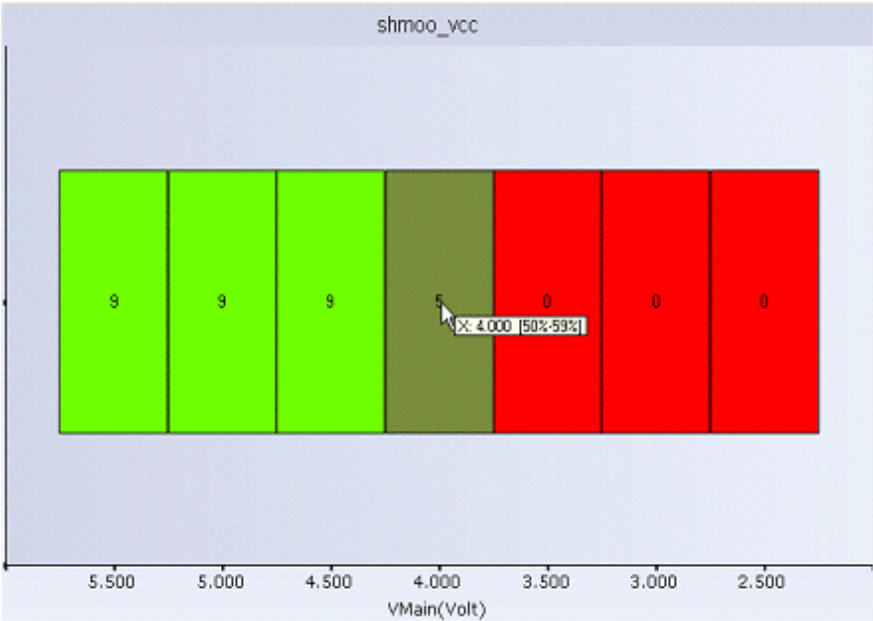
If a Shmoo or Margin involves multiple sites, pins, or patterns, instead of viewing the displays one at a time using the [Output Display Control](#), you can merge them into a single display. When displays can be merged, the Merge boxes in the Output display control are enabled:



Merge Boxes in the Output Display

The boxes that are enabled indicate that the setup contains data for multiple sites, pins, or patterns that can be merged into a single display. You can perform any combination of merges. In the box shown above, you can merge sites, pins, or sites and pins.

When Merge is enabled, cells that have both passing and failing results use a mix of green and red to indicate the relative number of passing and failing results. Each color represents a range in the percentage of passing results: the key is shown if the legend is enabled. A number in the cells indicates the percentage of the points that passed. (If the percentage is 0% passed, the symbol in the cell is #; if it is 100% passed, the symbol is \$.) Tooltips display the point values and the percentage range in which the cell falls.



Shmoo Plot Showing a Merge Cell

Note that many operations available on an unmerged Shmoo are unavailable on a merged display. You cannot zoom, rerun a selected range or a point, or display detailed point information. You can show or hide the legend and swap axes.

To unmerge the display, uncheck the boxes.

Merging a display does not rerun the setup. It is simply a different way of viewing results. In this way, a merge is different from an accumulated composite display, which actually discards the results of individual runs while accumulating the composite results. See Accumulating a Composite Shmoo.

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Characterization Studio : Using the General Tab

Using the General Tab

The top-level tab for each setup contains fields that apply to the entire setup:

General

☒ On sheet

Test method

Retest

Function name

Output format

Legacy

Function arguments

Interpose functions

Type	Function	Arguments
Pre Setup	whatdaif	presetup
Post Setup	whatdaif	postsetup

Output destinations

☒ Datalog

☐ Text file

☒ Sheet

☐ Output window

☐ Immediate window

Shmoo2

Margin Parameters

Contexts:

Patterns:

Instances:

Context Tool

General Tab

Most of these parameters apply to all characterization modes. Those that are specific to a mode are noted.

On sheet

This read-only box is checked if the setup is defined on a Characterization sheet. If it is not checked, it is defined in VBT. If you create a setup in Characterization Studio, this box not checked unless you save it to a Characterization sheet.

Test method	Specifies how the characterization is to repeat the test execution: Retest, Reburst, or Reburst Serial, or Run Function. See Test Method . Margin also supports Run Pattern. See Margin .
Output format	Controls how much information characterization keeps for each tested point and what data is available for output: • Legacy: The same output behavior as IG-XL before V7.10.00_uflx. • Enhanced: Collect and display additional information about each characterization step. • Verbose: Collect and display additional information about each point (or pattern execution). You can use VBT to modify these built-in formats and to create new ones.
Function name, Function arguments	Enabled when Test method is Run Function. See Run Function .
Output destinations	Controls where output is sent. You can specify any number of output destinations, including none. Destinations are: • Datalog: The current datalog destinations, as specified by DataCollect Setup or its VBT equivalents. Destinations can be the OutputWin window or a text file. Datalogging must be turned on for this option to have an effect. • Text file: A text file. You can enter a filename or browse to a filename. • Sheet: A test program worksheet. You can enter a sheet name. • Output window: The Test Program Output window. • Immediate window: The Visual Basic Editor Immediate window.

Interpose functions	This lets you specify interpose functions that are to be executed before and after the entire setup is executed: Pre Setup and Post Setup. Other interpose functions can be specified at the appropriate levels of the setup definition. See Interpose Functions.
Margin Parameters	These parameters are specific to Margin. See ADD.

Test Method

This drop-down lets you select how the characterization is to repeat the test execution: Retest, Reburst, or Reburst Serial, or Run Function.

Margin also supports Run Pattern. [See Margin.](#)

A multistep setup can have only one test method that applies to all of its steps.

☒ **Note:**

Most Teradyne digital instruments support the Reburst test method. SB6G does not. When characterizing a standalone SB6G pattern, you must use the Retest test method.

The Retest option re-executes the Body section of the test.

Reburst and Reburst Serial both reburst the most recently executed pattern. They differ in how they reburst the modules in a pattern set. The effect depends on whether the modules in the pattern set are to be executed all in a single burst or are to be burst separately:

- If modules in a pattern set are burst separately, then only the last module in the set is reburst, generating only one Shmoo based on that last module in the set. In this case, Reburst and Reburst Serial have the same effect.
- If modules in a pattern set are executed in a single burst, then Reburst and Reburst Serial have different results:
 - Reburst rebursts the pattern set in a single burst, generating a single Shmoo using all modules in the pattern set.
 - Reburst Serial rebursts each module in the set separately, in a serial loop, generating a separate Shmoo for each module in the pattern set.

Run Function

Run Function runs the user-created function for each point. If you select Run Function, the fields Function name and Function arguments are enabled. You enter the name of the function and

comma-separated list of any arguments. The VBT object TheExec.DevChar has statements that let you access information contained in the current characterization setup.

For more information, see [DevChar \(Device Characterization\)](#) in the VBT Syntax Reference.

Interpose Functions

This tab lets you specify interpose functions that are to be executed before and after the entire setup is executed: Pre Setup and Post Setup. Other interpose functions can be specified at the appropriate levels of the setup definition.

For more information, see [Using Interpose Functions](#) in the section on the Characterization sheet.

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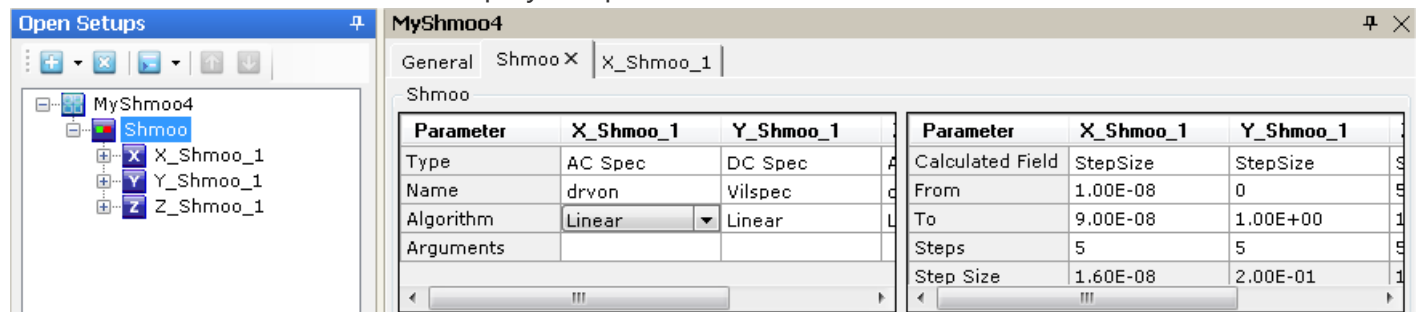
Characterization Studio : Shmoo

Shmoo

This section describes the parameters for setting up a shmoo. On the Shmoo output, see [Shmoo Display](#)

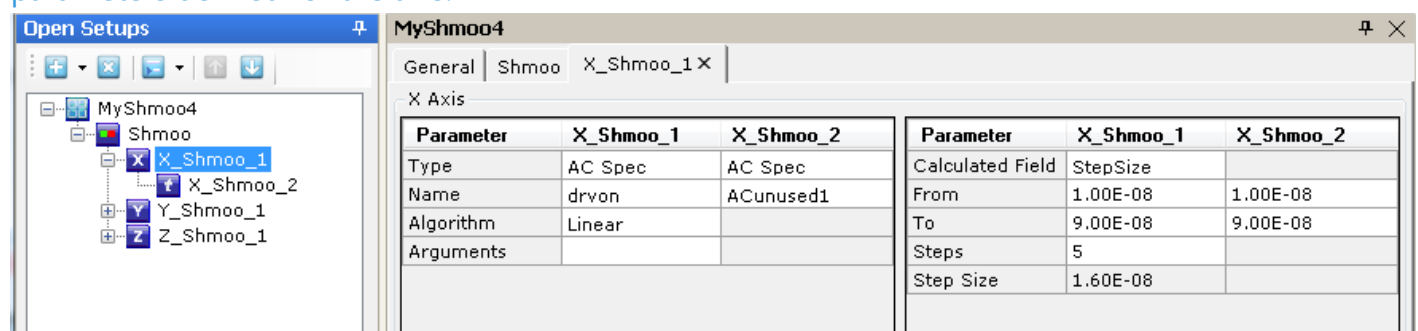
A Shmoo's steps are its X, Y, and Z axes. In addition, a Shmoo axis can have any number of tracking parameters.

The second-level Shmoo node displays all parameters for the axes:



Second-Level Parameters

Each third-level axis node repeats the parameters for that axis, and also shows any tracking parameters defined for the axis:



Third-Level Parameters

In addition, the third-level tab has controls for specifying Pins, TimeSets, and Interpose. A fourth-level node is used for each tracking parameter; it also has the controls for specifying pins, timesets, and interpose functions.

The following fields appear on the second-level Shmoo node and the third-level axis nodes:

Type	Select the type of parameter to vary: • AC Spec: Defined on an AC Specs sheet • DC Spec: Defined on a DC Specs sheet • Global Spec: Defined on the Global Specs sheet • Level: DC Level defined on a Pin Levels sheet • Edge: One of the timing edges (On, Data, Return, etc.) • Period: Period value defined on a Time Sets sheet • Protocol Aware: One of these values: Clock Offset, Drive Delay, ReceiveDelay, HiZ Delay, Reference Offset. • Serial Timing: Timing set defined on a Serial Timing sheet See Type Field.
Name	Select the name of the parameter to vary. The drop-down contains the names of the specified Type. If the type is Period, no name is used.
Algorithm, Arguments	Select the algorithm to be used to locate the Pass/Fail point. If the algorithm you select has arguments, enter them as a comma-separated list into the Arguments field. See Algorithm Field .
Calculated Field	Select the range parameter whose value is to be calculated using the other three range parameter values. The range field you select here will be read-only. You must provide values for the other three range fields.
From, To, Steps, Step Size	Enter the values to be used to define the range of the sweep. The value specified as Calculated Field is calculated from the other three range parameter values and is therefore read-only.

	The To value can be less than the From value. Reversed limits cause a backwards search over the search range.
Pin Execution Mode	Second-level Shmoo node only. Individual axes can specify that the characterization is to be applied only to certain pins or pin groups. If you use this option, you must specify whether to characterize pins simultaneously or individually. For a Shmoo, there is one setting for all axes. See Pins, TimeSets, and Interpose.
Interpose functions	Second-level Shmoo node only. Specify the Pre Step and Post Step interpose functions that are to be executed before and after each axis. The third-level axis nodes have controls for specifying Pre Point and Post Point functions. See Pins, TimeSets, and Interpose.

A tracking parameter has these fields:

•	Type
•	Name
•	From
•	To

In addition, it can specify pins, timesets, and Pre Point and Post Point interpose functions.

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Characterization Studio : Shmoo Display

Shmoo Display

This section describes the Shmoo display, particularly operations you can perform on the display. The following topics are available:

- | | |
|---|--|
| • | Shmoo Basics |
| • | Viewing the Third Axis |
| • | Shmoo Display Commands and Buttons |
| • | Merging Displayed Data |
| • | Accumulating a Composite Shmoo |
| • | Zoom |



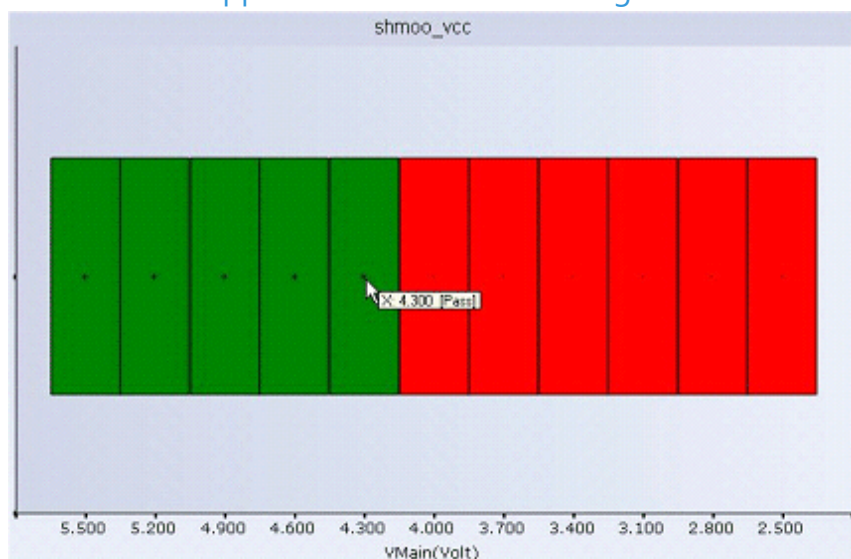
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Characterization Studio : Shmoo Display : Shmoo Basics

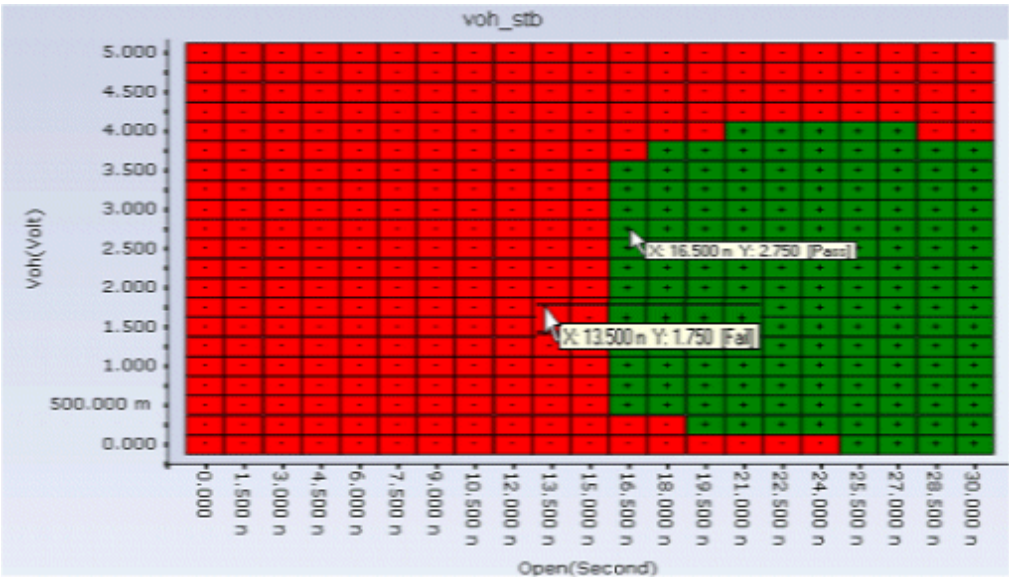
Shmoo Basics

A 1-D Shmoo appears as a line of cells along the X axis:



1-D Shmoo Plot

A 2-D Shmoo appears as a grid of square cells:



2-D Shmoo Plot

A 3-D Shmoo looks like a 2-D Shmoo, with controls available for viewing the third axis. See [Viewing the Third Axis](#).
The results of the test at each point are indicated by the color and in some cases the symbol of the cell:

- Green and + = Pass
- Red and - = Fail
- Gray = No test
- White = No test result, or unable to test

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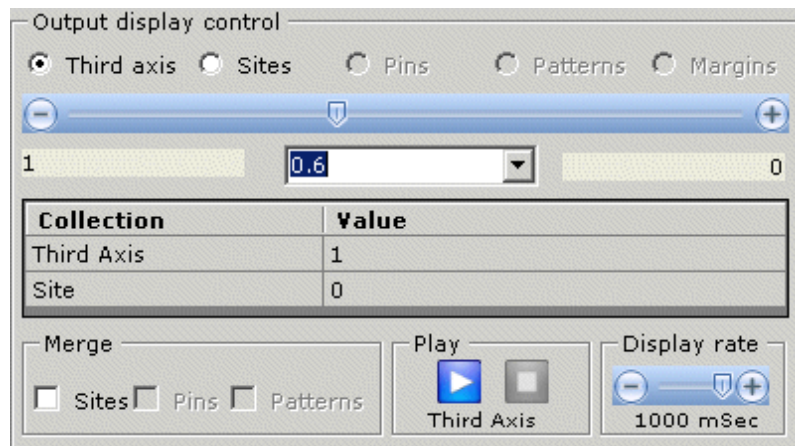


Characterization Studio : Shmoo Display : Viewing the Third Axis

Viewing the Third Axis

Characterization Studio displays a 3-D Shmoo in two dimensions. To view the third axis, use the Output control display. For more information, see [Output Display Control](#).

The control looks like this:



Third Axis in the Output Control Display

Select the radio button labeled Third Axis. The Output display control now contains the steps of the third axis. It has the number of Steps in the setup, going from the value of the From step to the value of the To step. You can move the slider to the value you want, select the value from the drop-down, or play the slideshow to display each step. In this way, you can navigate the third axis, viewing it slice by slice.

You can also select which of the three axes to display in the two dimensions. By default, the display includes the X and Y axes, and you use the Third Axis display to show the Z axis.

You can change the axes that are displayed in the two dimensions. See the [Axes](#) button and command, which swap the two specified axes. Third Axis always refers to the axis not currently on the two-dimensional display. If you have swapped the Z and A axis, so that the display currently shows the Y and Z axes, the Third Axis control shows the values of the X axis.

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Characterization Studio : Shmoo Display : Shmoo Display Commands and Buttons

Shmoo Display Commands and Buttons

The display area has toolbar buttons, menu commands, and a right-click context menu commands that let you control the output display. This section lists and describes operations that are available for the Shmoo display.

Toolbar

Re-Run, Stop, Zoom, Revert, Refresh, Axes, plus the standard Copy, Print, and Save buttons

Menu Bar

The menus on the display area menu bar, from left to right, are:

Edit	Standard Cut, Copy, and Paste commands.
Output Display	Re-Run, Revert, Refresh, Zoom
Axes	See Axes. Enabled only for Shmoo.

Context Menu

When you right-click in the Shmoo display, a context menu becomes available with these options, from top to bottom (note that some commands on the context menu have names different from those on the menu bar):

Launch PatternTool, Show Detailed Point Info, Re-Run Selected, Zoom Selected, Run Point and Restore, Show/Hide Legend

Axes



Axes menu

Swaps the axes currently displayed Shmoo. Swapping axes does not cause the setup to be rerun. It simply displays the results in a different way.

The button and menu have a drop-down that lists the possible swaps.

For a 2-D Shmoo, only Swap XAxis - YAxis is enabled. You can swap the positions of the two axes. For a 3-D Shmoo, you can swap one of the two currently displayed axes for the undisplayed axis. That is, if the X and Y axes are displayed, you can replace either with the Z axis. For a 3-D Shmoo, all swaps are enabled on the drop-down list, with these control sequence equivalents:

- Swap XAxis - YAxis - Ctrl+Shift+X

- Swap YAxis - ZAxis - Ctrl+Shift+Y

- Swap ZAxis - AAxis - Ctrl+Shift+Z

On using Axes to display the Z axis, see [Viewing the Third Axis](#).

Launch PatternTool

Context menu only

Brings up PatternTool with the test’s pattern open.

Show Detailed Point Info

Context menu only

Brings up a window that displays detailed information about the selected Shmoo point:

Detail Info For Point X: 0.6 Y: 1.42857142857143E-08

General Info

Site	0
X Index	3
Y Index	3
Z Index	N/A
X	600.000 m
Y	14.286 n
Z	N/A
Result	No Test

Tracking Parameter Details

Axis	Parameter	Value
X	X_Shmoo_vih	0.4

Adjust Parameter Details

Step Name	Parameter	Test Result	PGA Value
vdd_min_a	vdd	NoTest	2.99

Errors and Alarms

Errors:/Alarms:

Detailed Point Info Display

The information includes information about the point location and values; the setup parameters; the pattern and the first failing cycle, module, pin, and vector, when available; details about any tracking parameters; details about any Adjusts or Measures included in the setup; and any errors or alarms.

Refresh

 Refresh Output Display>Refresh, Ctrl+F

Refreshes the display with current data from memory.

Re-Run

 Re-Run Output Display>Re-Run, Ctrl+R, Context menu>Re-Run Selected

Rerun only the selected range, leaving the rest of the Shmoo display unchanged. Before executing this command, first select a range of cells in the display by dragging across them. The zoom and axis orientation remain unchanged.

You usually rerun the selected range after modifying a setup.

Revert

 Revert Output Display>Revert, Ctrl+T

After running a zoom, the Revert button and command are available to let you restore the original Shmoo display. Revert does not rerun the setup. See [Zoom](#).

Run Point and Restore

Context menu only

Run the Shmoo point, then restore the hardware’s timing and levels settings. Rerunning a point usually leaves the hardware set to the parameters of the point. Run Point and Restore runs the point but leaves the hardware in the default condition for the test context rather than for the point.

Show/Hide Legend

Context menu only

Show or hide the explanation of the colors of the boxes in the display.

Stop

 Stop While a Re-Run or Zoom is running, this button is enabled to stop re-execution of the setup.

Zoom

 Zoom Output Display>Zoom, Ctrl+O, Context menu>Zoom Selected

Rerun the selected range and zoom in on it to display it in greater detail. See [Zoom](#).

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Characterization Studio : Shmoo Display : Accumulating a Composite Shmoo

Accumulating a Composite Shmoo

You can run a Shmoo multiple times and accumulate the results. If you select a Shmoo setup, the numeric control to the left of the Runs button is enabled:



Runs Button with Numeric Control Enabled

You increment the number to indicate the number of times you want to run the Shmoo. The resulting display is a composite of the multiple executions. The display is similar to a merged display, including the use of shaded colors to represent the percentage of passes and fails at each point. See [Merging Displayed Data](#). The difference is that, after running each of the multiple executions, the data from the run is added to the accumulated image and is then discarded. You cannot view the results of a single run. In contrast, a merge is a way of viewing multiple data sets at once. Results are not actually merged, and you can unmerge the display by unchecking the Merge box.

An accumulated composite Shmoo does not allow you to rerun the selected range or zoom.

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Characterization Studio : Shmoo Display : Zoom

Zoom

You can zoom in on a selected range of Shmoo points to see the points in greater detail. First select the portion of the Shmoo by dragging across the range. Then use one of these methods:

-  Click the Zoom button on the display area toolbar.
- Right-click the display and select Zoom from the context menu.

The selected portion of the Shmoo is rerun using the same number of steps as the original run, thereby showing the selected portion in greater resolution.

You can zoom in further on an already zoomed Shmoo.

After you perform a zoom, the Revert button is enabled. Revert restores the original display. It does not rerun the characterization.

When you have zoomed in on a Shmoo, the zoomed version becomes the stored results. If you later use Open Results From Memory for the setup, if the zoomed version is the most recently executed, it is the one that is displayed. See [Opening Stored Results From Memory](#).



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Characterization Studio : Margin

Margin

A Margin can contain one or more individual Margins as steps. The following describes how to set up and display a Margin:

- Margin General Tab
- Running Patterns in Contexts
- Margin and Step Tabs
- Margin Display

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Characterization Studio : Margin : Margin General Tab

Margin General Tab

For a Margin, the General tab has all the parameters available for the other modes: see [Using the General Tab](#). In addition, it has a set of Margin-specific parameters and a Context Tool:

General

☒ On sheet

Test method

Run Pattern

Function name

Output format

Legacy

Function arguments

Interpose functions

Type	Function	Arguments
Pre Setup		
Post Setup		

Output destinations

☒ Datalog

☐ Text file

☐ Sheet

☐ Output window

☐ Immediate window

Margin Parameters

Contexts:

Patterns: NR_low, NR_high

Instances:

Context Tool

Margin General Tab

Margin Parameters are enabled based on the Test method selected:

- If the test method Run Pattern is selected, the Margin parameter Instances is disabled, and Contexts, Patterns, and Context Tool are enabled. See [Running Patterns in Contexts](#).
- If any test method except Run Pattern is selected, the Margin parameter Instances is enabled, and Contexts, Patterns, and Context Tool are disabled. This field is ignored when you are running a setup from Characterization Studio. It is used only when you are characterizing the flow using VBT. See [Running Margins on a Flow from VBT](#).

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Characterization Studio : [Margin](#) : Running Patterns in Contexts

Running Patterns in Contexts

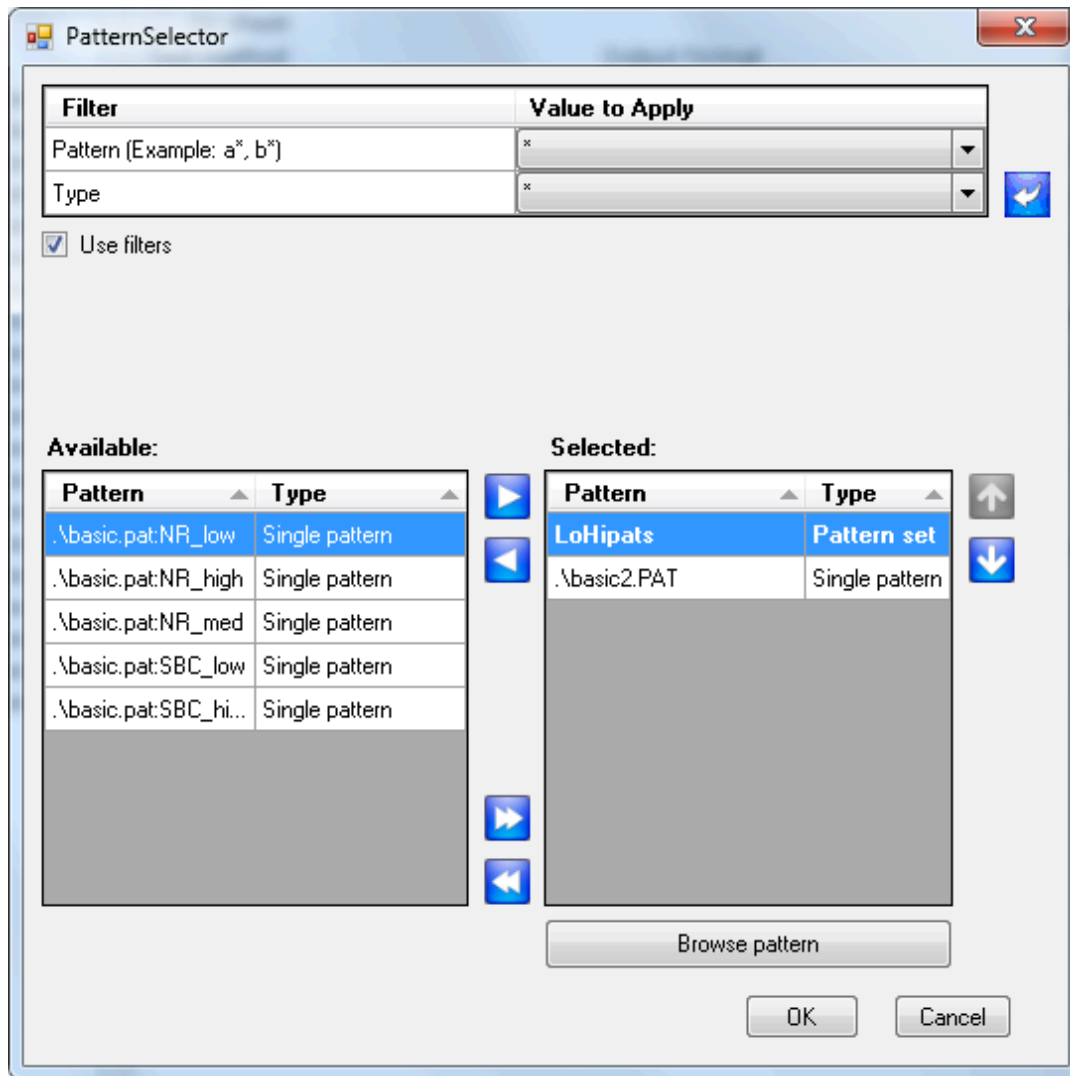
If the test method is Run Pattern, the Patterns and Contexts parameters are enabled, so that you can specify which patterns to run and which contexts are to be used to run them. This lets you run a pattern using a context other than the one specified in the test instance that runs the pattern.

You can select the context to be used from existing contexts, or you can use Context Tool to specify a new context.

The Margin is run once for each specified pattern in each specified context. You can specify multiple patterns and contexts.

Specifying the Patterns

To specify the patterns, click the ellipsis button next to the Patterns field on the General tab. This brings up the PatternSelector:



PatternSelector

By default, the Available window lists the patterns that have been loaded. You can use the Browse pattern button to select patterns on disk that have not been loaded. The windows indicate whether an item is a single pattern (which includes a single module within a pattern) or a pattern set. You can filter the list of patterns by name, using a wildcard template, or by type, selecting single pattern or pattern set from the drop-down list.

Specifying the Contexts

By default, a context in IG-XL is not an independent named unit. Instead, a context is defined as a set of parameters for a test instance: the AC and DC category and selector, and the timing, pin levels, and mixed signal timing sheets.

Instead of directly selecting an existing context to use in a setup, you select a test instance that defines the context. The rest of the test instance parameters are ignored: only the context parameters are used for the setup.

You can select a test instance other than the one that actually runs the pattern in the job. This lets you run a pattern in different contexts.

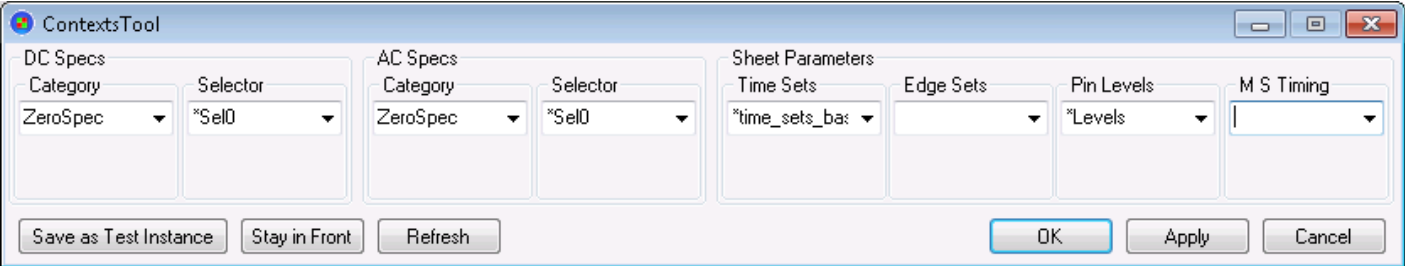
To specify the contexts, click the ellipsis button next to the Contexts field on the General tab. This brings up the ContextSelector. The Available window on the selector lists the test instances currently

in use in the flow. You can filter on the test instance names with a wildcard template to reduce the number of instances listed.

Select the test instances that define the contexts you want to use and move then to the Selected window. Then click OK.

Using Context Tool

Instead of using an existing context defined by a test instance, you can create a new context using the Context Tool. Click the Context Tool button to bring up the tool:



Context Tool

The Context Tool button is disabled if any contexts are currently specified in the Contexts field.

The tool initially displays the values for the current context. An asterisk indicates that the currently displayed value is the initial value. You can select different values from the drop-down for each field.

When you click OK or Apply, you apply the currently specified values as the context in which the Margin is to be run. You can also save the currently displayed values as a named test instance. You specify the name of the new instance, which is saved to the Test Instances sheet.

Once you select the values you want, you can use the buttons to perform these actions:

Context Tool Buttons

Button	Action
Save as Test Instance	Save the currently specified values as a test instance. You are prompted for a name.
Stay in Front	Keep Context Tool in front of other tools. This can be useful if you are make several context changes and running the Margin after each change.
Refresh	Restore the initial value of any context parameter that has not been applied yet. Once you click Apply, the changed value is stored.
OK	Apply all changes and exit from Context Tool.
Apply	Apply all changes, but do not exit from Context Tool.
Cancel	Exit from Context Tool without applying changes.

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Characterization Studio : Margin : Margin and Step Tabs

Margin and Step Tabs

The [second-level](#) Margin node displays all parameters for the individual steps. Each third-level step node repeats the parameters for that step; in addition, it has controls for specifying [Pins](#), [TimeSets](#), and [Interpose](#).

The following fields appear on the [second-level](#) Margin node and the third-level step nodes:

Type	Select the type of parameter to vary:
	• AC Spec: Defined on an AC Specs sheet • DC Spec: Defined on a DC Specs sheet • Global Spec: Defined on the Global Specs sheet • Level: DC Level defined on a Pin Levels sheet • Edge: One of the timing edges (On, Data, Return, etc.) • Period: Period value defined on a Time Sets sheet • Protocol Aware: One of these values: Clock Offset, Drive Delay, ReceiveDelay, HiZ Delay, Reference Offset.

- [Serial Timing](#): Timing set defined on a Serial Timing sheet

See [Type Field](#).

Name	Select the name or names of the parameter to vary. Click the ellipsis button to bring up the SpecSelector, which functions like the other selector. The Available window lists all specs of the selected Type. You can filter the names using a wildcard template. If you select multiple names, they appear on the Margin tab as a comma-separated list. If the type is Period, no name is used, and the SpecSelector button is disabled.
Algorithm, Arguments	Select the algorithm to be used to locate the Pass/Fail point. If the algorithm you select has arguments, enter them as a comma-separated list into the Arguments field. See Algorithm Field.
Calculated Field	Select the range parameter whose value is to be calculated using the other three range parameter values. The range field you select here will be read-only. You must provide values for the other three range fields.
From, To, Steps, Step Size	Enter the values to be used to define the range of the sweep. The value specified as Calculated Field is calculated from the other three range parameter values and is therefore read-only. The To value can be less than the From value. Reversed limits cause a backwards search over the search range. If you are characterizing an AC or DC spec, the From and To fields have a drop-down that includes the strings Min, Typ, and Max. These represent the current values for the specified spec from the AC or DC Specs sheet.

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Characterization Studio : Margin : Margin Display

Margin Display

For a Margin, the display shows the Margins in the setup. For each Margin, the individual steps of the Margin execution are shown between the From and To points, with colors indicating passing and failing steps. Numeric results are displayed in a grid.

The only menu and toolbar command is Refresh. Use this command to refresh the display with current data from memory.

The only commands enabled on the right-click context menu are:

Launch PatternTool	Brings up PatternTool with the pattern open.
Show/Hide Legend	Shows or hides the explanation of the colors of the boxes in the display.

You can use the Output display control to view the results along any third dimension in the setup. For example, the following setup, which has multiple steps and multiple pins, also specifies multiple patterns and contexts and is run multisite:

☒ On sheet

Test method

Run Pattern

Function name

Output format

Legacy

Function arguments

Type	Function	Arguments
Pre Setup		
Post Setup		

Output destinations

☒ Datalog

☐ Text file

☐ Sheet

☐ Output window

☐ Immediate window

Margin Parameters

Contexts: VBFailSite3,VBPassstoFail

Patterns: LoHipats,.\basic2.PAT

Instances:

Context Tool

Margin Setup

In the output display control area, the radio buttons lets you specify the third dimension to be displayed:

Output display control

Third axis

Sites

Pins

Patterns

Margins

Instances

Contexts

LoHipats

LoHipats

.\basic2.PAT

Merge

Sites

Pins

Patterns

Instances

Contexts

Play

Patterns

Display rate

1600 mSec

Margin Output

For more information, see [Output Display Control](#).

You can also merge the output data along one or more of the third dimensions. Check the boxes in the Merge area. See [Merging Displayed Data](#).

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Characterization Studio : Adjust, Adjust From, and Measure

Adjust, Adjust From, and Measure

Adjust and Measure have the same parameters, except that an Adjust can write the measured value to a spec, with an optional backoff value.

The [second-level characterization mode nodes](#), Adjusts and Measures, display the parameters for all of a setup's steps. Each third-level step node repeats the parameters for the specific step and also has controls for specifying [Pins](#), [TimeSets](#), and [Interpose](#).

Type

Select the type of parameter to vary:

- **AC Spec**: Defined on an AC Specs sheet
- **DC Spec**: Defined on a DC Specs sheet
- **Global Spec**: Defined on the Global Specs sheet
- **Level**: Dc Level defined on a Pin Levels sheet
- **Edge**: One of the timing edges (On, Data, Return, etc.)
- **Period**: Period value defined on a Time Sets sheet
- **Serial Timing**: Timing set defined on a Serial Timing sheet

See Type Field.

Name

Select the name of the parameter to vary. The drop-down contains the names of the specified Type. If the type is Period, no name is used.

**Algorithm,
Arguments**

Select the algorithm to be used to locate the Pass/Fail point. If the algorithm you select has arguments, enter them as a comma-separated list into the Arguments field. See Algorithm Field.

Results Check

Select which check to perform:

- **Disabled**: Do not perform any check.
- **First**: Check the first point in the range.
- **Last**: Check the last point in the range.
- **Both**: Check both points in the range.

A results check examines one or both endpoints of the search range before running the characterization to make sure that the point matches what is expected based on the specified transition.

For example, if the transition is PassToFail, the First results check makes sure that the first point in the range actually passes. If it does not, the characterization is not run because it could not produce the desired transition, and it returns a “Stuck at Fail” error code.

Results Check cannot be First if Transition is also First.

Transition

Select which transition the characterization should attempt to locate:

- **First:** Find the first transition, either pass-to-fail or fail-to-pass.
- **PassToFail:** Find only a pass-to-fail transition.
- **FailToPass:** Find only a fail-to-pass transition.

If you select PassToFail and FailToPass, you can also request a results check before running the characterization.

Spec Name

Adjust only. Optionally select from the drop-down list the name of the spec that will receive the results of the Adjust. This can be the spec being varied or a different spec. See Performing Adjust and Measure.

Backoff

Adjust only. Optionally enter the value to be applied to the adjusted value. A negative backoff value therefore makes the result smaller. A backoff value is applied to the result to leave the test passing with a certain safety margin. See Performing Adjust and Measure.

Comment

Enter a comment. This is stored in the Comment column of the Characterization sheet.

Calculated Field

Select the range parameter whose value is to be calculated using the other three range parameter values. The range field you select here will be read-only. You must provide values for the other three range fields.

From, To, Steps, Step Size

Enter the values to be used to define the range of the sweep.

The value specified as Calculated Field is calculated from the other three range parameter values and is therefore read-only.

The To value can be less than the From value. Reversed limits cause a backwards search over the search range.

Perform Test, Test Limit Low, Test Limit High

Specify whether to execute the limits test using the limits given in Test Limits Low and Test Limits High. If the indicated transition does not occur within the specified limits, the test is datalogged as a Fail.

Adjust From

Adjust From has the same parameters on its second-level characterization mode node and its third-level step node.

From Setup	Select the name of the setup from which the adjusted spec value is to be retrieved.
Spec Name	Select the name of the levels or timing spec whose value is to be taken from the named setup.
Comment	Enter a comment. This is stored in the Comment column of the Characterization sheet.

Adjust and Measure Display

For an Adjust or Measure, the display consists of a simple grid with the results. The only menu and toolbar command is Refresh. Use this command to refresh the display with current data from memory. The only command on the right-click context menu is Launch PatternTool, which brings up PatternTool with the test's pattern open. You can use the Output display control to view the results along the dimension of sites, pins, or patterns. See Output Display Control. For the other parts of the display, see Output Display Area: Overview.

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Characterization Studio : General Notes on Mode-Specific Parameters

General Notes on Mode-Specific Parameters

The following sections apply to these parameters on the mode-specific tabs:

- Pins, TimeSets, and Interpose
- Type Field
- Algorithm Field

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Characterization Studio : General Notes on Mode-Specific Parameters : Pins, TimeSets, and Interpose

Pins, TimeSets, and Interpose

If you select the third-level step node or the fourth level Shmoo tracking parameter node, you can access the controls for specifying pins, timesets, and interpose functions. For instructions on using these controls, see [Specifying Pins, TimeSets, and Interpose Functions](#).

For Measure and Margin, if you specify pins, you must also select a value from the Pin Execution Mode drop-down, to specify whether to characterize the pins simultaneously or individually. The values are:

Simultaneous	Apply the search conditions to all pins at the same time and report a single value.
PinByPin	Decompose any pin groups into individual pins, and then vary the pins one at a time, reporting a separate result for each pin. Note that differential pins are not supported in PinByPin mode, because the differential pair is decomposed into single-ended pins. When using extended I/O plus dual-compare mode on the HPM-XIO instrument only, pairs of alternating adjacent channels starting with channels 0 and 1, then 4 and 5, and so on share the VOH and VOL level DACs. Since you cannot adjust VOH or VOL independently on these adjacent channel pairs, IG-XL returns incorrect values when running PinByPin characterization on VOH or VOL levels for pins that are normally drive-only pins when not in extended I/O mode. For standard I/O channels that do not share VOH and VOL level DACs (that is, channels 2 and 3, then 6 and 7, and so on), this restriction does not apply.
ItemByItem	Do not decompose pin groups into individual pins. Instead, vary the pin items one pin or pin group at a time, reporting a separate result for each

pin or pin group.

For a Shmoo, all axes use the same pin execution mode, so the control is on the second-level Shmoo tab.

For Adjust, Measure, and Margin, you can specify the following interpose functions for each step:

- Pre Step and Post Step, to be executed before and after executing each step
- Pre Point and Post Point, to be executed before and after performing the test at each point

For Shmoo, you specify Pre Step and Post Step on the second-level Shmoo node, and Pre Point and Post Point on the third-level axis nodes and the fourth-level tracking parameter nodes.

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Characterization Studio : [General Notes on Mode-Specific Parameters](#) : Type Field

Type Field

You can vary a Level or Edge parameter on a pin only if the pin is already programmed to a state that is compatible with the parameter being varied. For example, you can vary the Close edge of a pin if the pin is already in window mode and has a Close edge programmed. If the pin is in edge mode (with an Open edge but no Close edge), specifying that the Close edge is to be varied does not change the pin from edge to window mode. Similarly, if an edge has the value Disabled, specifying that the edge is to be varied does not enable the edge.

You cannot vary a Spec parameter if it is used in the definition of one of these Pin Levels parameters: Tdelay, VLevel, or VPpmu. If a spec is used to define one of these levels, varying the spec causes a run-time error.

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Characterization Studio : General Notes on Mode-Specific Parameters : Algorithm Field

Algorithm Field

You select the algorithm to be used to generate the values for the characterization. The algorithms are:

Binary	Generate values by continuously selecting the midpoint of the defined range (From and To) and updating the endpoints of the range. The algorithm continues updating the endpoints until the range is equal to or less than the step size (Steps and Step Size). This algorithm produces values that are integer increments of the step size, with values rounded up if necessary.
Edge	Do two linear searches, from the From and To points. Each linear search stops at the first test result change. Shmoo and Margin only.
Jump	Do a linear search that starts with the first point and then skips the number of points specified in the Arguments column. Ignore any From, To, Steps, and Step Size values.
Linear	Generate values from From to To in fixed steps calculated from Steps and Step Size. This algorithm produces values up to and including the end of the range but will not exceed the end of the range.
Successive	Generate values using a combination of the binary and linear search algorithms. A binary search is used until a transition is found between two consecutive points, and then a linear search is used between these points.

	The Successive algorithm is intended to be used with characterization of a response that drifts with time, such as chasing a moving edge. Adjust and Measure only.
List	Use a specific set of values provided in a comma-separated list in the Arguments field. Ignore any From, To, Steps, and Step Size values.

Note that HPM and UltraPin4000 have a restriction on algorithms that you can use to characterize drive edges of site-copied channels simultaneously, that is, in parallel. This situation can occur when a pin on the channel map is assigned site-copied channels on different sites. For this characterization, you can use only the Linear or List algorithm. Any other algorithm may fail to find the transition point.

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CharacterizationDefTopic

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About Characterization

About Characterization

This document covers IG-XL characterization. It has the following sections:

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| • | Device Characterization |
| • | Characterization Studio |
| • | Other Characterization Features |

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